

INTEGRATING GITHUB & RSTUDIO WORKSHOP

Topics covered: version control, installation and setup, basic commands, and practical application

Bring a laptop, some patience, and a friend!



Hochstetter 549
September 8
1:30 PM EST
Or join on Teams!



BEFORE WE BEGIN

- URLs and Links to click on will be in [light blue](#)
- Code will be in `orange Fira Code`

CAVEATS

- You should have some familiarity with RStudio
- Very basic – we are covering the bare-minimum you need to use GitHub with Rstudio
- I do not know everything! I am only sharing what works for me
- We are going to go SLOW
- Almost all material on this presentation is borrowed from the book HappyGitwithR.com

Happy Git and GitHub
for the useR

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Let's Git started

1 Why Git? Why GitHub?

2 Contributors

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Installation

Half the battle

4 Register a GitHub account

5 Install or upgrade R and RStudio

6 Install Git

7 Introduce yourself to Git

8 Install a Git client

Connect Git, GitHub, RStudio

Can you hear me now?

9 Personal access token for

Let's Git started



Still from Heaven King video



AGENDA

VERSION CONTROL

GIT VS GITHUB

PAIN + SUFFERING (INSTALLATION)

HI GIT, I'M RSTUDIO

COMMON RSTUDIO STRUCTURES

WHAT NEXT

GIT VS GITHUB



- Software foundation of Git version control
- Open source
- ~95% percent of developers use some version



- Web hosting service for Git repositories (repos)
- Most popular online Git service
- Subsidiary of Microsoft

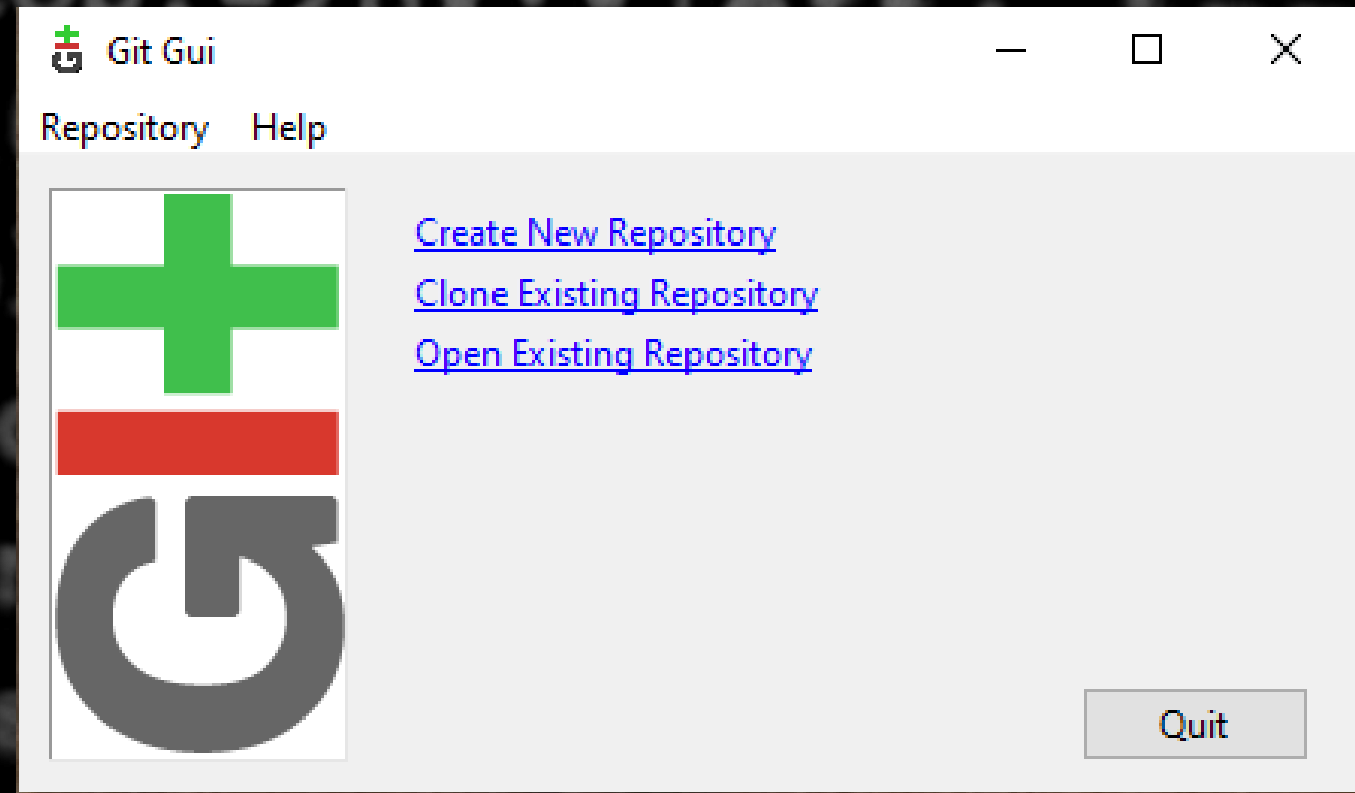
LET'S TALK ABOUT VERSION CONTROL

Name	Status	Date modified	Type	Size
 Coauthor Code		7/8/2025 1:06 PM	File folder	
 Coauthor Revisions		7/8/2025 1:06 PM	File folder	
 Figures and Tables		7/8/2025 1:06 PM	File folder	
 First Submission (rejected)		7/8/2025 1:06 PM	File folder	
 OrganizedDec2023LitSearchResults		7/8/2025 1:06 PM	File folder	
 Reviewer Comments		7/8/2025 1:06 PM	File folder	
 Second Submission		8/6/2025 10:30 AM	File folder	
 Supplemental Materials		7/24/2025 5:51 PM	File folder	
 Fish Assemblage and Climate Change v18		3/14/2025 1:49 PM	Microsoft Word D...	280 KB
 Second Submission Fish Assemblage and Climate Change V1		4/25/2025 4:05 PM	Microsoft Word D...	279 KB
 Second Submission Fish Assemblage and Climate Change V2		4/25/2025 7:53 PM	Microsoft Word D...	274 KB
 Second Submission Fish Assemblage and Climate Change V3_Writing Club		4/25/2025 7:23 PM	Microsoft Word D...	270 KB
 Second Submission Fish Assemblage and Climate Change V4		5/1/2025 8:27 PM	Microsoft Word D...	271 KB
 Second Submission Fish Assemblage and Climate Change V5		5/9/2025 3:54 PM	Microsoft Word D...	278 KB
 Second Submission Fish Assemblage and Climate Change V6		5/15/2025 4:35 PM	Microsoft Word D...	289 KB
 Second Submission Fish Assemblage and Climate Change V7		5/27/2025 3:01 PM	Microsoft Word D...	286 KB
 Second Submission Fish Assemblage and Climate Change V8		5/28/2025 2:25 PM	Microsoft Word D...	287 KB
 ShoemakerResponseLetterV1		5/9/2025 3:34 PM	Microsoft Word D...	35 KB
 ShoemakerResponseLetterV2		5/15/2025 4:54 PM	Microsoft Word D...	36 KB
 ShoemakerResponseLetterV3		5/23/2025 2:25 PM	Microsoft Word D...	32 KB
 UTF-8ShoemakerResponseLetterV1		5/2/2025 5:28 PM	Microsoft Word D...	32 KB

Name	Date modified	Type	Size
 Final Revisions Thesis Draft	2/11/2021 3:56 PM	Microsoft Word D...	8,017 KB
 Shoemaker_Darren Thesis Submitted Version	5/25/2021 6:07 PM	Firefox PDF Docu...	721 KB
 These Draft Template Update	11/19/2020 9:54 PM	Microsoft Word D...	24,592 KB
 These Final Draft for Formatting Meeting 3_6_2021 Tables on Own Pages	3/11/2021 5:11 PM	Microsoft Word D...	7,736 KB
 These Final Draft for Formatting Meeting 3_6_2021 Tables on Same Page	3/8/2021 12:32 PM	Microsoft Word D...	7,733 KB
 These Final Draft for Formatting Meeting 3_6_2021	3/6/2021 5:11 PM	Microsoft Word D...	7,734 KB
 Thesis Draft All Revisions	2/3/2021 11:37 PM	Microsoft Word D...	8,005 KB
 Thesis Draft Before Formatting Meeting	2/18/2021 4:57 PM	Microsoft Word D...	8,012 KB
 Thesis Draft Discussion	11/28/2020 3:44 PM	Microsoft Word D...	23,736 KB
 Thesis Draft Gillespie February Meeting	2/10/2021 12:01 PM	Microsoft Word D...	8,017 KB
 Thesis Draft Marshall Revisions	2/2/2021 8:10 PM	Microsoft Word D...	8,625 KB
 Thesis Draft Revised Discussion	2/26/2021 3:27 PM	Microsoft Word D...	8,012 KB
 Thesis Draft Rocky Revisions	1/28/2021 3:02 PM	Microsoft Word D...	8,615 KB
 Thesis Draft with Presence Models	1/27/2021 2:19 PM	Microsoft Word D...	24,013 KB
 Thesis Final Draft for Rocky	2/26/2021 3:28 PM	Microsoft Word D...	8,011 KB

WHY GIT?

- Add structure to version control
- Maintain versions across devices and collaborators
- Recover old versions
- Share analyses/projects with a single link



```

01_range.R x  lulc x
Source on Save Run Source
1 #' ---
2 #' title: "Hellbender Range
3 #' author: "Darren Shoemaker"
4 #' ---
5
6 # Install Packages
7
8 #remotes::install_github("rspatial/luna")
9
10 # Libraries
11
12 library(terra)
13 library(tidyterra)
14 library(sf)
15
16 library(ggplot2)
17 library(ggspatial)
18
19 library(here)
20
21 # Read Data
22
23 state <- map_data('state')
24
25 hell <- read_sf(here('data/aHELLa_CONUS_Range_2001v1',
26 'aHELLa_CONUS_Range_2001v1.shp'))
27
28 iucn <- read_sf(here('data/iucn', 'data_0.shp'))
29
30 extent <- ext(xmin = 242666.578099998, xmax = 1773643.625, ymin =
31 1236427.5, ymax = 2403071.75)
32
33 # Stream Geodatabase
34
35 st_layers('E:/R_Stuff/02_big_data/NHDPlus_H_National_Release_2_GDB
36 .gdb')
37
38 stream <- st_read('E:/R_Stuff/02_big_data
39 /NHDPlus_H_National_Release_2_GDB.gdb', options = '--config
40 OGR_SKIP_OpenFileGDB', layer = 'NetworkNHDFlowline')
41
42 # Earth Explorer
43
44 elev <- getNASA(product = 'SRTMGLOBAL1N', path = 'E:/R_Stuff
45 /02_big_data/', username = 'dshoemaker', password = 'Sjc9_mFG"XC
46 ,h3C')

```

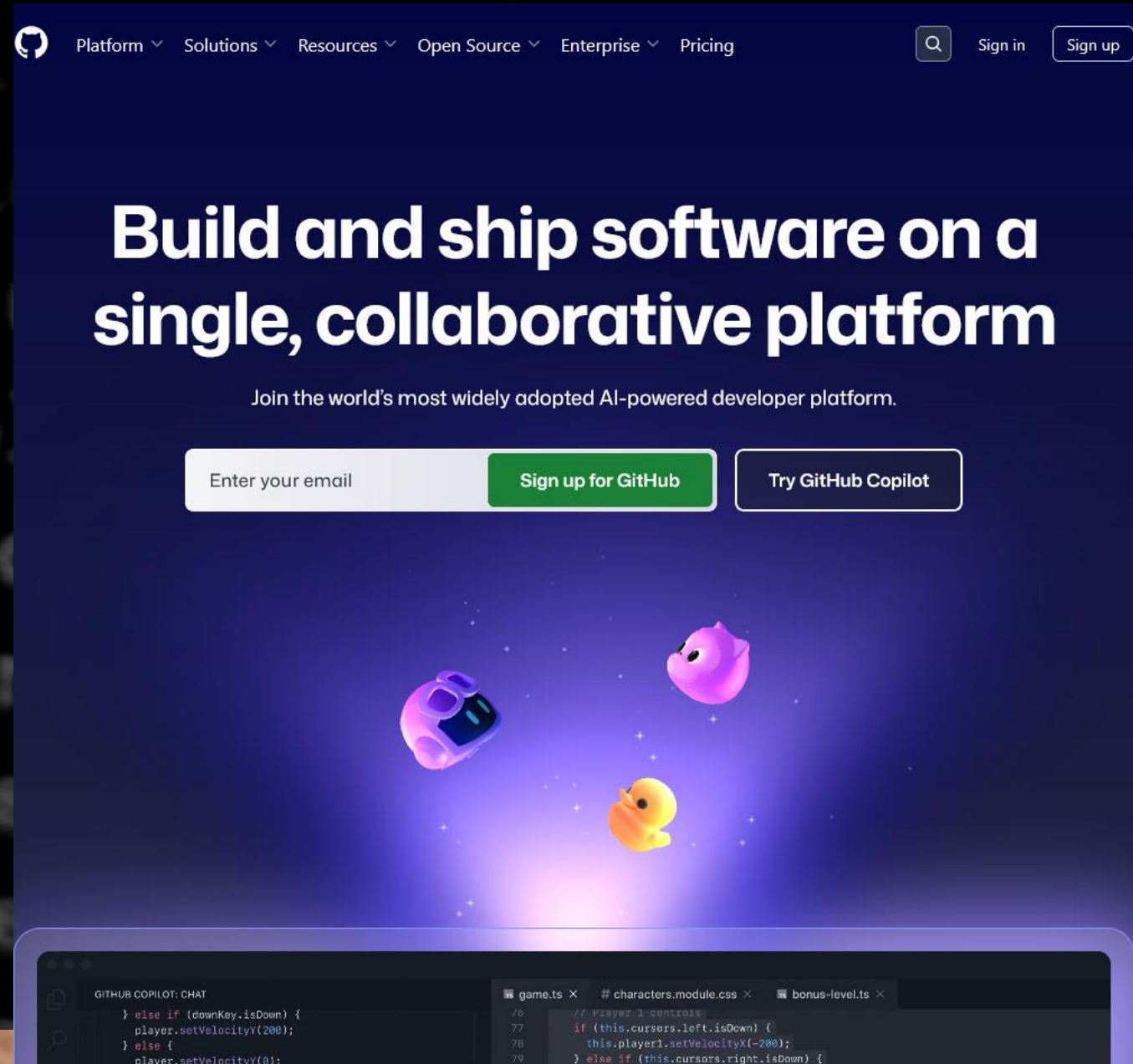
WHY GITHUB?

- Widely adopted
- “Easy” integration with RStudio
- Lots of support and resources

MAKE AN ACCOUNT

Be thoughtful with your username!

- You *can* change this later, but it will hurt
- Use an email that you'll have for awhile
- Use only lowercase letters and use underscores (_) to make spaces if necessary
- Future employers *will* see your account
- DO NOT PAY FOR ANYTHING!



CHECK YOUR RSTUDIO VERSION

- Try to be no more than two minor versions behind
- Not the end of the world but can cause issues

```
> RStudio.Version()
$citation
To cite RStudio in publications use:

Posit team (2025). RStudio: Integrated
Development Environment for R. Posit
Software, PBC, Boston, MA. URL
http://www.posit.co/.
```

A BibTeX entry for LaTeX users is

```
@Manual{,
  title = {RStudio: Integrated Development Environment
for R},
  author = {{Posit team}},
  organization = {Posit Software, PBC},
  address = {Boston, MA},
  year = {2025},
  url = {http://www.posit.co/},
}
```

```
$mode
[1] "desktop"
```

```
$version
[1] '2025.5.0.496'
```

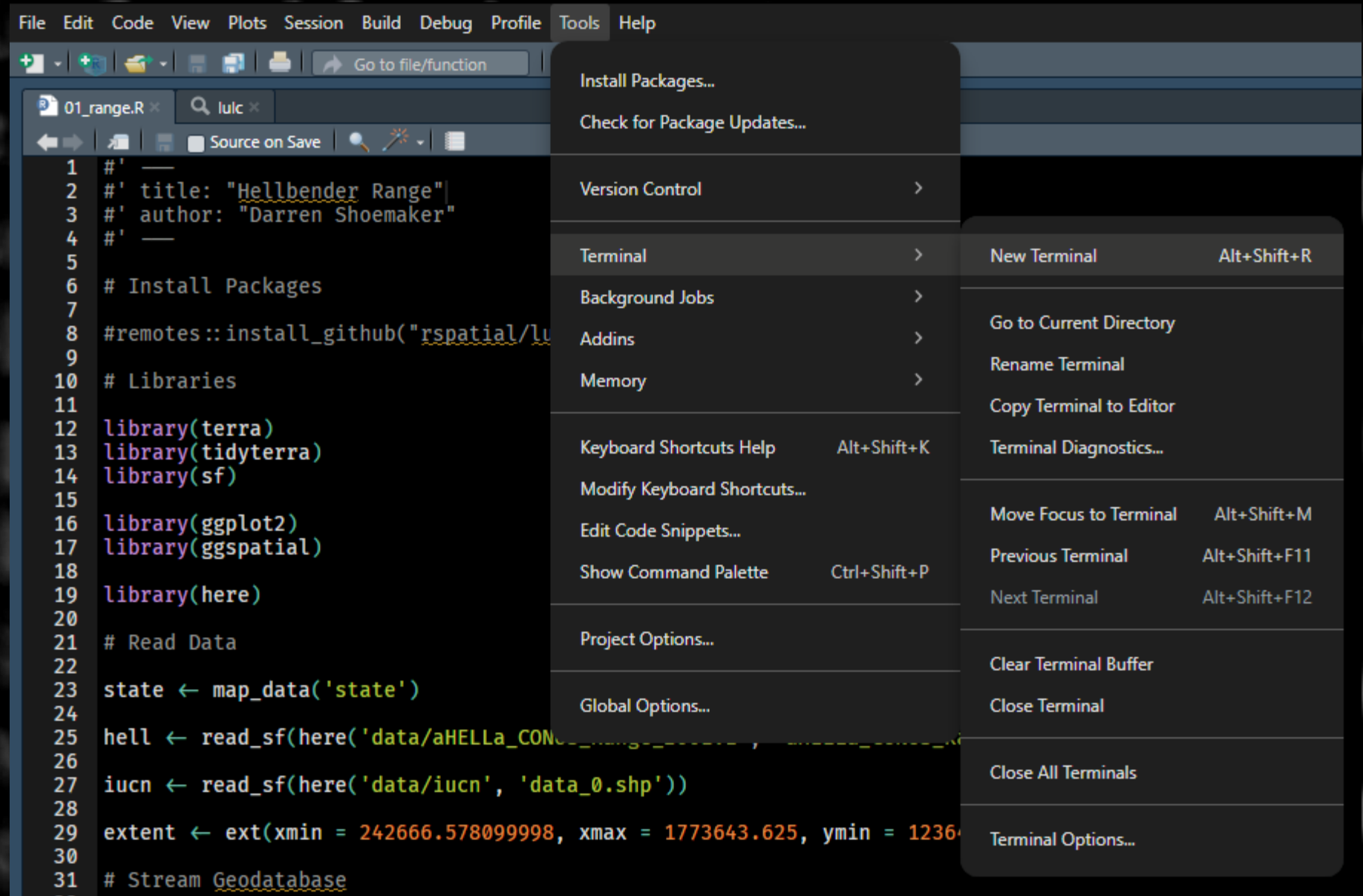
```
$long_version
[1] "2025.05.0+496"
```

```
$release_name
[1] "Mariposa Orchid"
```

```
> |
```

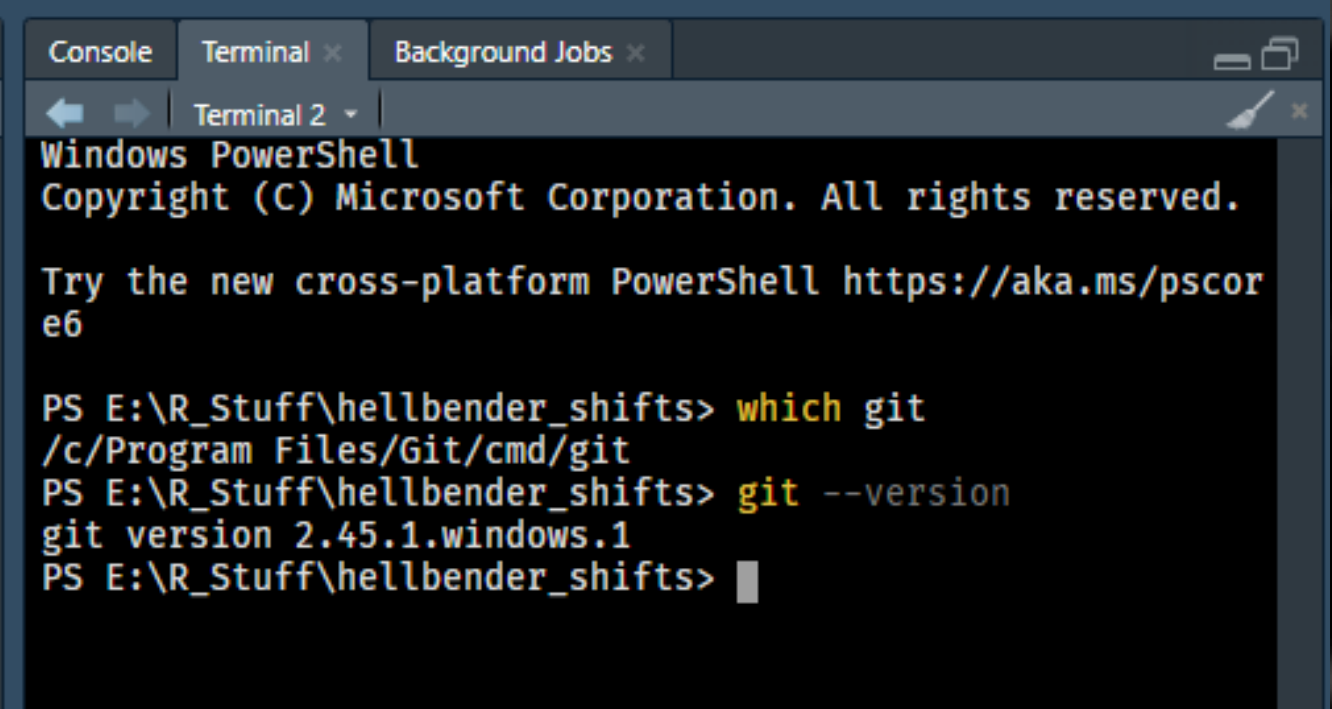

PAIN + SUFFERING

CHECK IF
GIT IS
ALREADY
INSTALLED



THE TERMINAL

- In the terminal, type
- “`which git`”
- If you get a file, type
- “`git -- version`”
- If you have version 2.4 or later, you can skip installation



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS E:\R_Stuff\hellbender_shifts> which git
/c/Program Files/Git/cmd/git
PS E:\R_Stuff\hellbender_shifts> git --version
git version 2.45.1.windows.1
PS E:\R_Stuff\hellbender_shifts>
```


▼ Assets 15

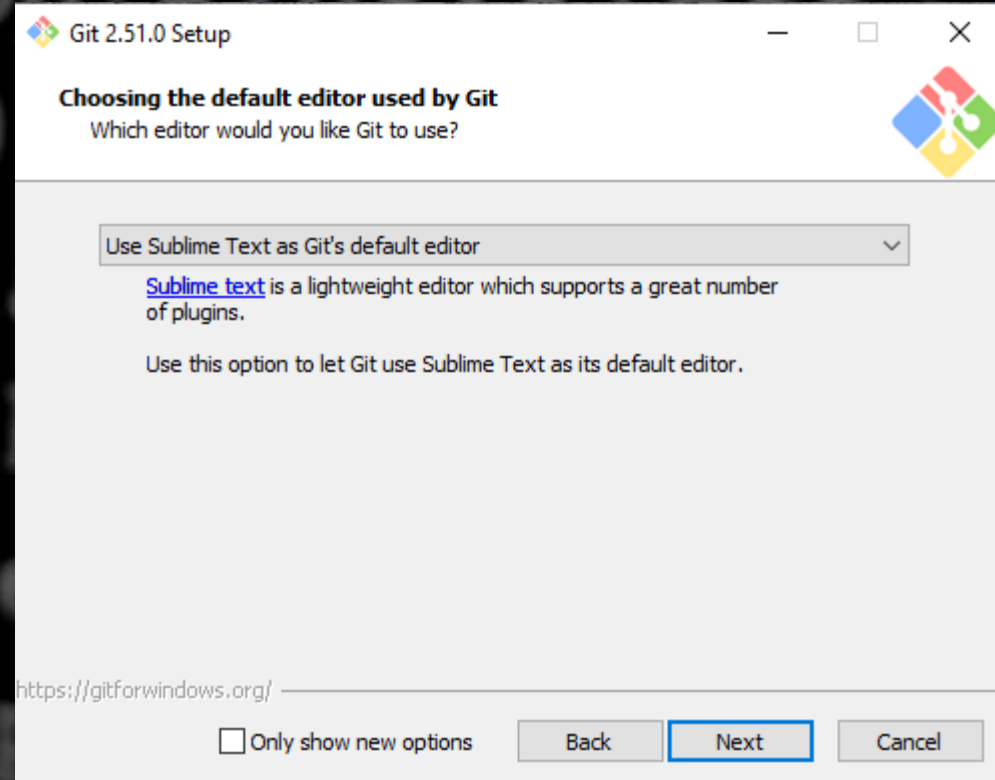
 Git-2.51.0-64-bit.exe	sha256:843037416371600a7f289be8fe2b2...		61.7 MB	last week
 Git-2.51.0-64-bit.tar.bz2	sha256:151bddf70e1115631e62bb05535b5...		110 MB	last week
 Git-2.51.0-arm64.exe	sha256:739673a52a2ea5a3ac23ef1a74985...		59.7 MB	last week
 Git-2.51.0-arm64.tar.bz2	sha256:5c3bc6ca50ef6a7686832d2549e6e...		148 MB	last week
 MinGit-2.51.0-32-bit.zip	sha256:5a8f1cace31a817fa9fa3d18146e8...		40 MB	last week
 MinGit-2.51.0-64-bit.zip	sha256:c2c955a21fa99889d83f485f24fa5...		39.4 MB	last week
 MinGit-2.51.0-arm64.zip	sha256:b21755ccd10f71a37ec341ca9ac45...		38.2 MB	last week
 MinGit-2.51.0-busybox-32-bit.zip	sha256:050fe76ece1b7762cd556bdb242a...		35.7 MB	last week
 MinGit-2.51.0-busybox-64-bit.zip	sha256:6b71de89d321310d1cc233565a10b...		35 MB	last week
 pdb-for-git-64-bit-2.51.0.1.4d21a77b98-1.zip	sha256:a4b08c041ed95d79bd67c869d664e...		16.5 MB	last week
 pdb-for-git-arm64-2.51.0.1.4d21a77b98-1.zip	sha256:29f8e7cfffdeb0bc76d457877243db...		2.2 MB	last week
 PortableGit-2.51.0-64-bit.7z.exe	sha256:a09b275d51ed3e829128e04cf4168...		56.6 MB	last week
 PortableGit-2.51.0-arm64.7z.exe	sha256:0aacd4edf0c1715334a18725a9475...		56.6 MB	last week
 Source code (zip)				last week
 Source code (tar.gz)				last week

  15  3  4  1 16 people reacted

 1 [Join discussion](#)

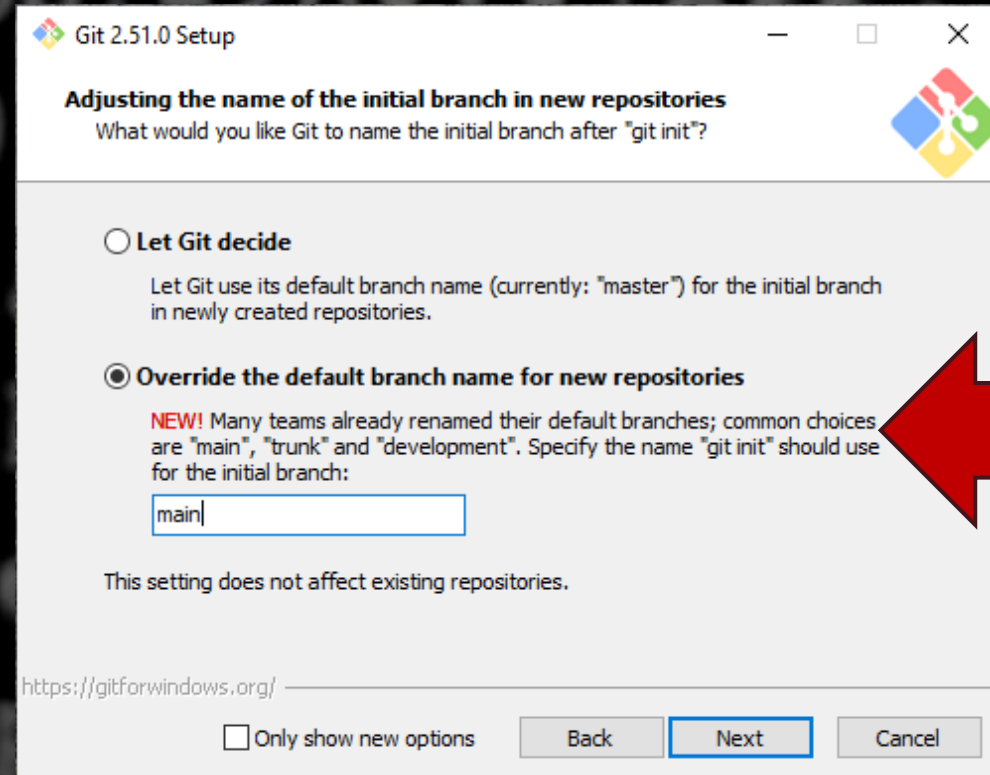
INSTALLATION SETTINGS

I don't know what this means, but
Sublime Text was recommended
to me and I have not had issues



INSTALLATION SETTINGS

Idk seems woke...



The image shows the 'Git 2.51.0 Setup' window. The title bar says 'Git 2.51.0 Setup'. The main heading is 'Adjusting the name of the initial branch in new repositories' with the subtext 'What would you like Git to name the initial branch after "git init"?'. There are two radio button options: 'Let Git decide' and 'Override the default branch name for new repositories'. The second option is selected. Below it, a note says 'NEW! Many teams already renamed their default branches; common choices are "main", "trunk" and "development". Specify the name "git init" should use for the initial branch:'. A text input field contains the word 'main'. At the bottom, there is a checkbox for 'Only show new options' and three buttons: 'Back', 'Next', and 'Cancel'. A large red arrow points from the right side of the screen towards the 'Override' option.

Git 2.51.0 Setup

Adjusting the name of the initial branch in new repositories
What would you like Git to name the initial branch after "git init"?

☐ Let Git decide
Let Git use its default branch name (currently: "master") for the initial branch in newly created repositories.

☒ **Override the default branch name for new repositories**
NEW! Many teams already renamed their default branches; common choices are "main", "trunk" and "development". Specify the name "git init" should use for the initial branch:

This setting does not affect existing repositories.

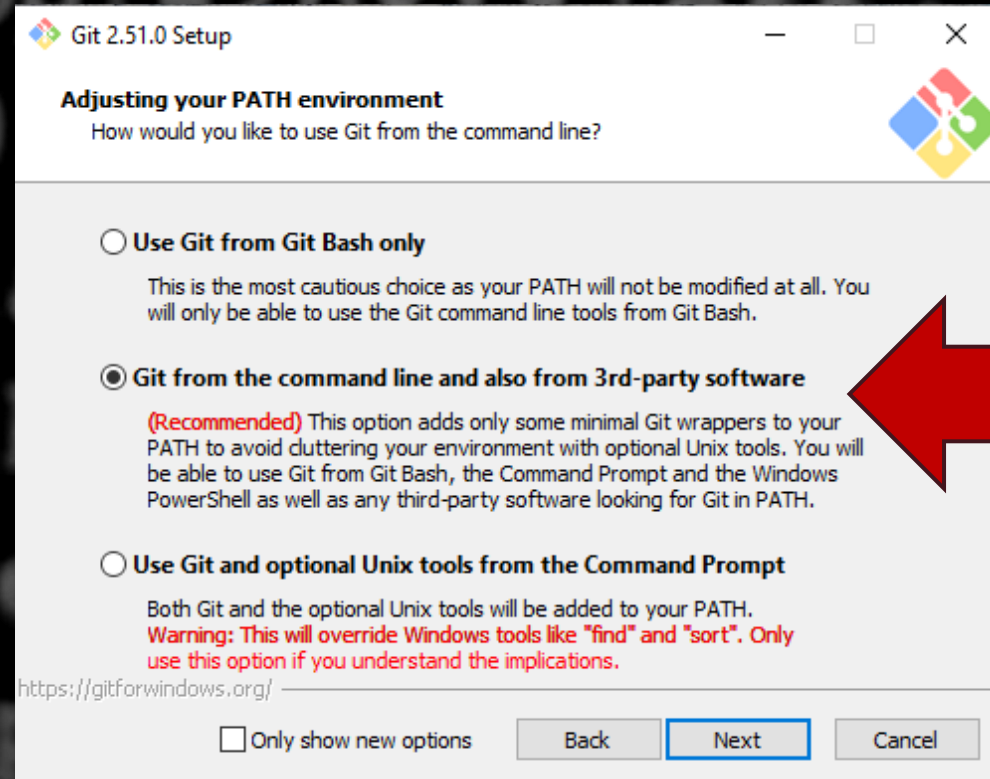
<https://gitforwindows.org/>

☐ Only show new options Back **Next** Cancel

INSTALLATION SETTINGS

Make sure you pick Git from the command line and also 3rd-party software

Leave everything else on default settings



TAKE A DEEP BREATH. THIS IS
ABOUT TO SUCK...

CREDENTIALS

- GitHub is strict about security
- We need to generate a personal access token (PAT) to give Rstudio “permission” to use GitHub
- We will use the HTTPS protocol. There are other options, but I haven’t tried them
- Go back to your GitHub account, go to settings, and click `<>` **Developer settings**
- This is way at the bottom for some reason...

Settings

Search Type to search

Go to your personal profile

Darren Shoemaker (thedungeonecologist)
Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Moderation

Code, planning, and automation

Repositories

Codespaces

Models

Packages

Copilot

Pages

Saved replies

Security

Code security

Integrations

Applications

Scheduled reminders

Archives

Security log

Sponsorship log

<> Developer settings

Public profile

Name

Darren Shoemaker

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your [email settings](#).

Bio

My name is Darren Shoemaker. I recently completed my Ph.D. at Mississippi State University examining climatic influences on ecological systems

You can @mention other users and organizations to link to them.

Pronouns

he/him

URL

Social accounts

<https://www.researchgate.net/profile/Darren-Shoemaker>

<https://www.linkedin.com/in/darrenshoemaker>

Link to social profile 3

Link to social profile 4

Company

You can @mention your company's GitHub organization to link it.

Location

Starkville

☐ Display current local time

Other users will see the time difference from their local time.

ORCID iD

You have a connected ORCID iD 0009-0002-3581-3944 for the account @thedungeonecologist.

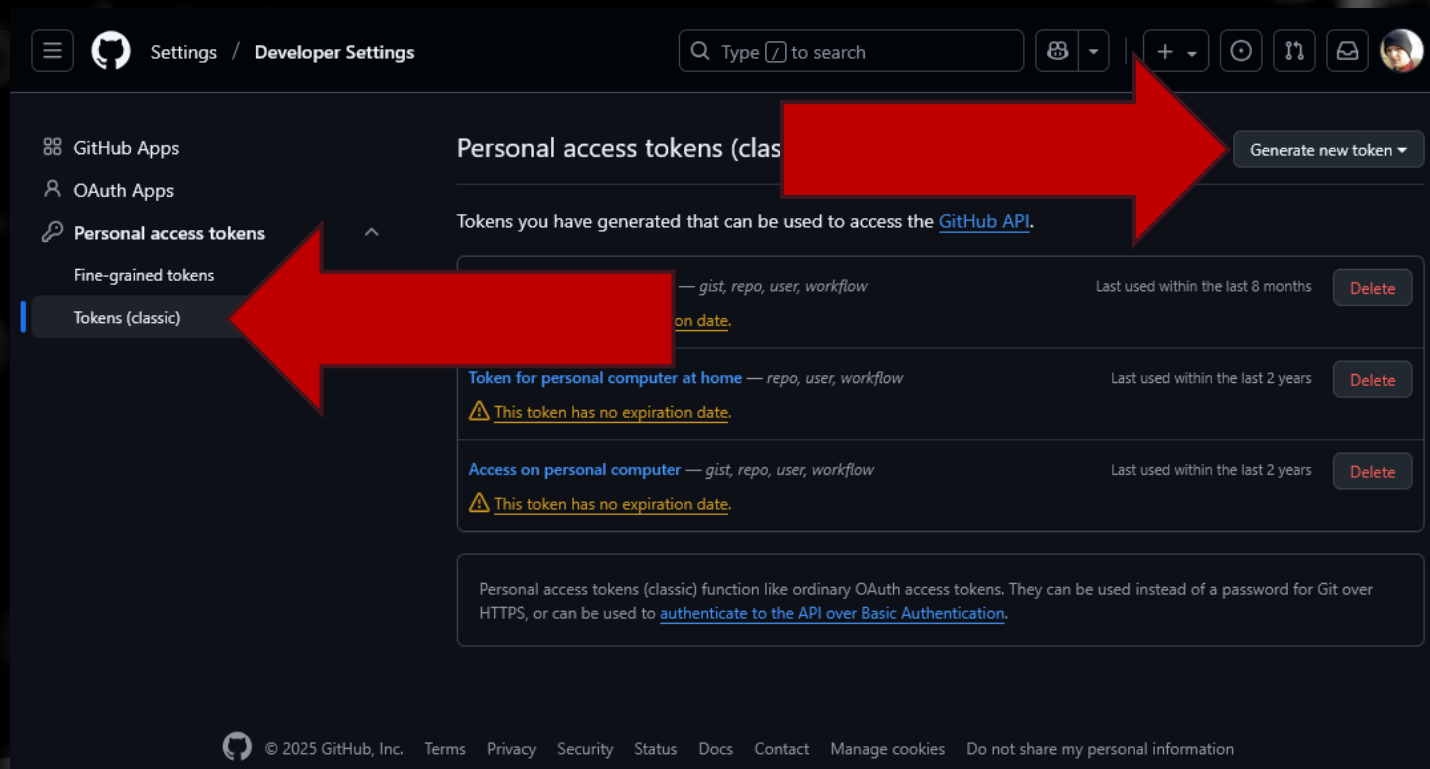
☒ Display your ORCID iD on your GitHub profile

Disconnecting your ORCID iD may affect areas of your profile where your ORCID iD is displayed.

Disconnect your ORCID iD

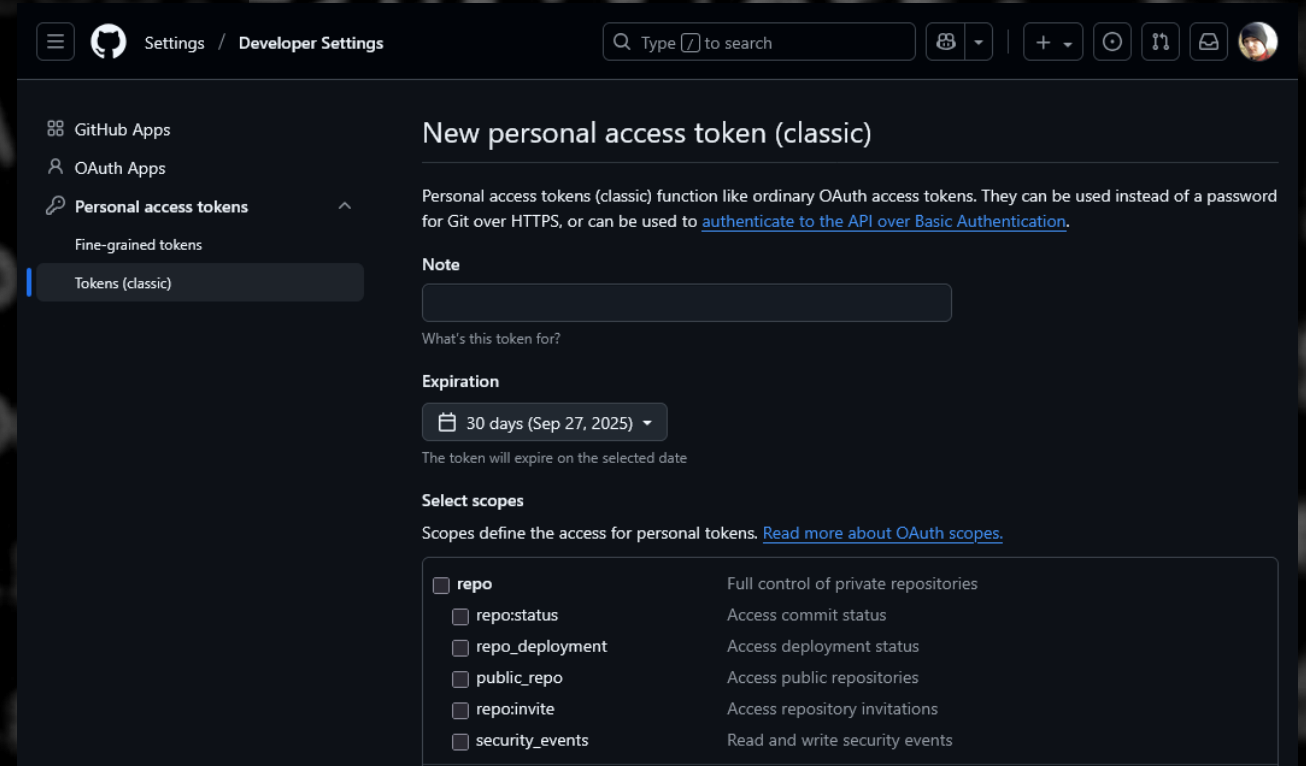
CREDENTIALS

- Click **Tokens (classic)**
- Fine-grained is repo-specific. This is more secure, but tedious
- Classic will give access to most of your repositories
- Click **Generate new token**



CREDENTIALS

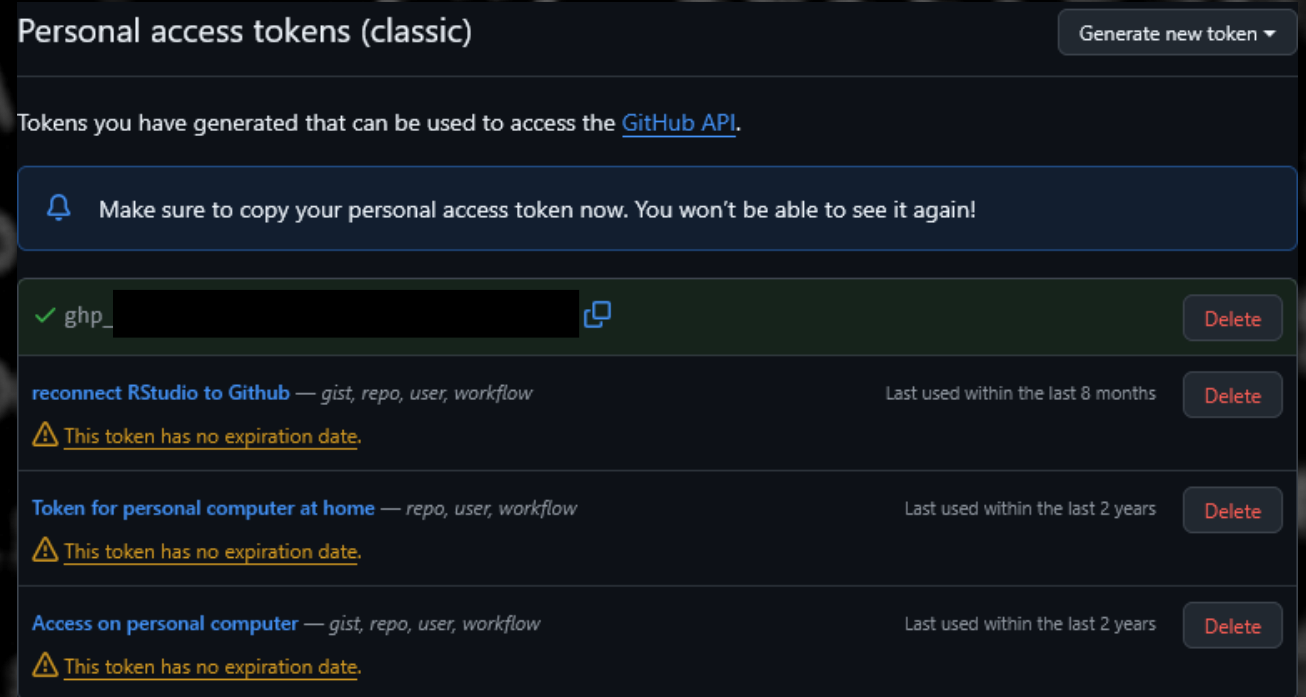
- I recommend checking **repo**, **user**, and **workflow**
- GitHub recommends updating new tokens every 30 days
- But wait, on the last slide yours had no expiration date?
- Use your own judgment on security. If you have sensitive data, let them expire
- I usually name the token after the device it's on. I have unique tokens for my office, home, and laptop



CREDENTIALS

DO NOT CLOSE THIS WINDOW UNTIL YOU SAVE THE TOKEN!

- GitHub recommends saving to a password management software
- Alternatively, you can copy it to the sticky notes app...
- Don't do that, but I have seen it
- A sticky note is probably fine for now

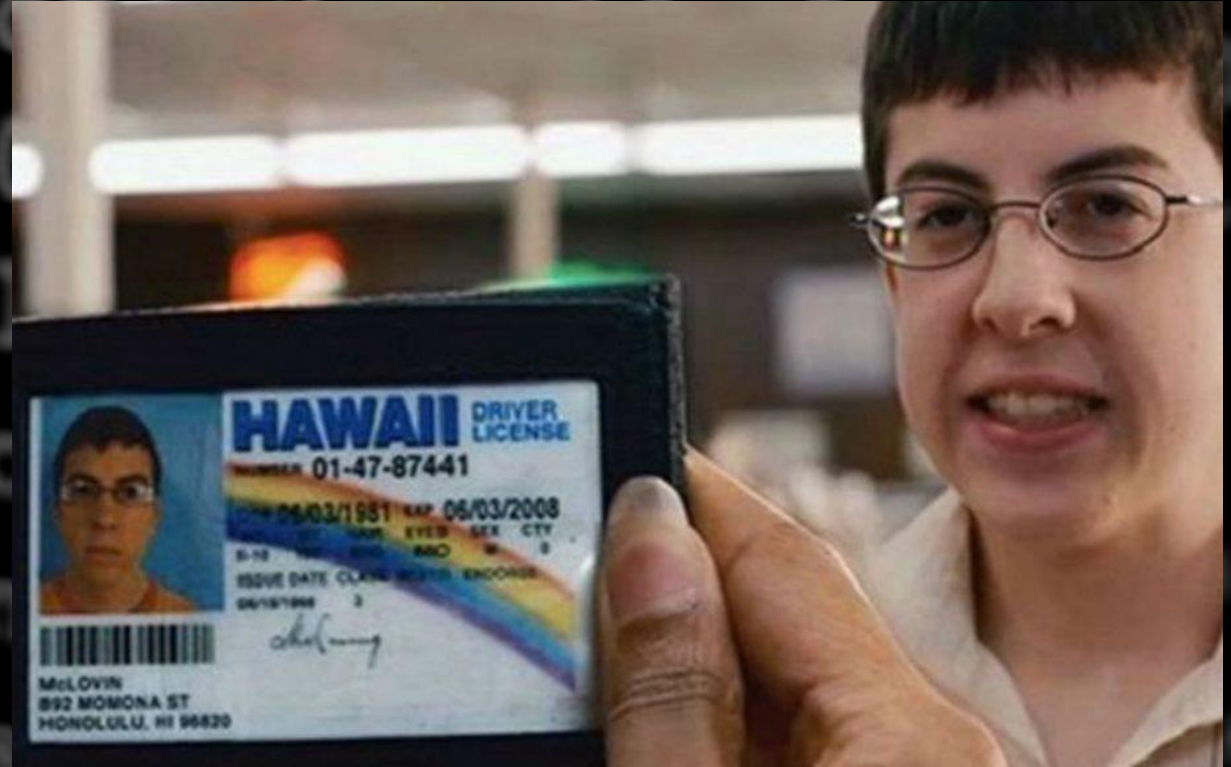


HI GIT, I'M
RSTUDIO!



SET YOUR CREDENTIALS

- Install the `gitcreds` and `usethis` R packages and load the libraries
- Enter `gitcreds::gitcreds_set()` in the console
- Paste your PAT when prompted
- You can check if this worked with `gitcreds::gitcreds_get()`

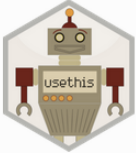


GITCREDS AND USETHIS

- From here on we're going to mostly work in the **gitcreds** and **usethis** R packages
- These packages are R wrappers around Git and lets us perform Git operations from the R console
- Everything here can also be done from the Terminal, but this is usually more convenient

usethis 3.2.0 Setup Reference Articles ▾ News ▾

Search for 



usethis

usethis is a workflow package: it automates repetitive tasks that arise during project setup and development, both for R packages and non-package projects.

Installation

Install the released version of usethis from CRAN:

```
install.packages("usethis")
```

Or install the development version from GitHub with:

```
# install.packages("pak")
pak::pak("r-lib/usethis")
```

LINKS

[View on CRAN](#)
[Browse source code](#)
[Report a bug](#)

LICENSE

[Full license](#)
[MIT](#) + file [LICENSE](#)

COMMUNITY

[Contributing guide](#)
[Code of conduct](#)
[Getting help](#)

CITATION

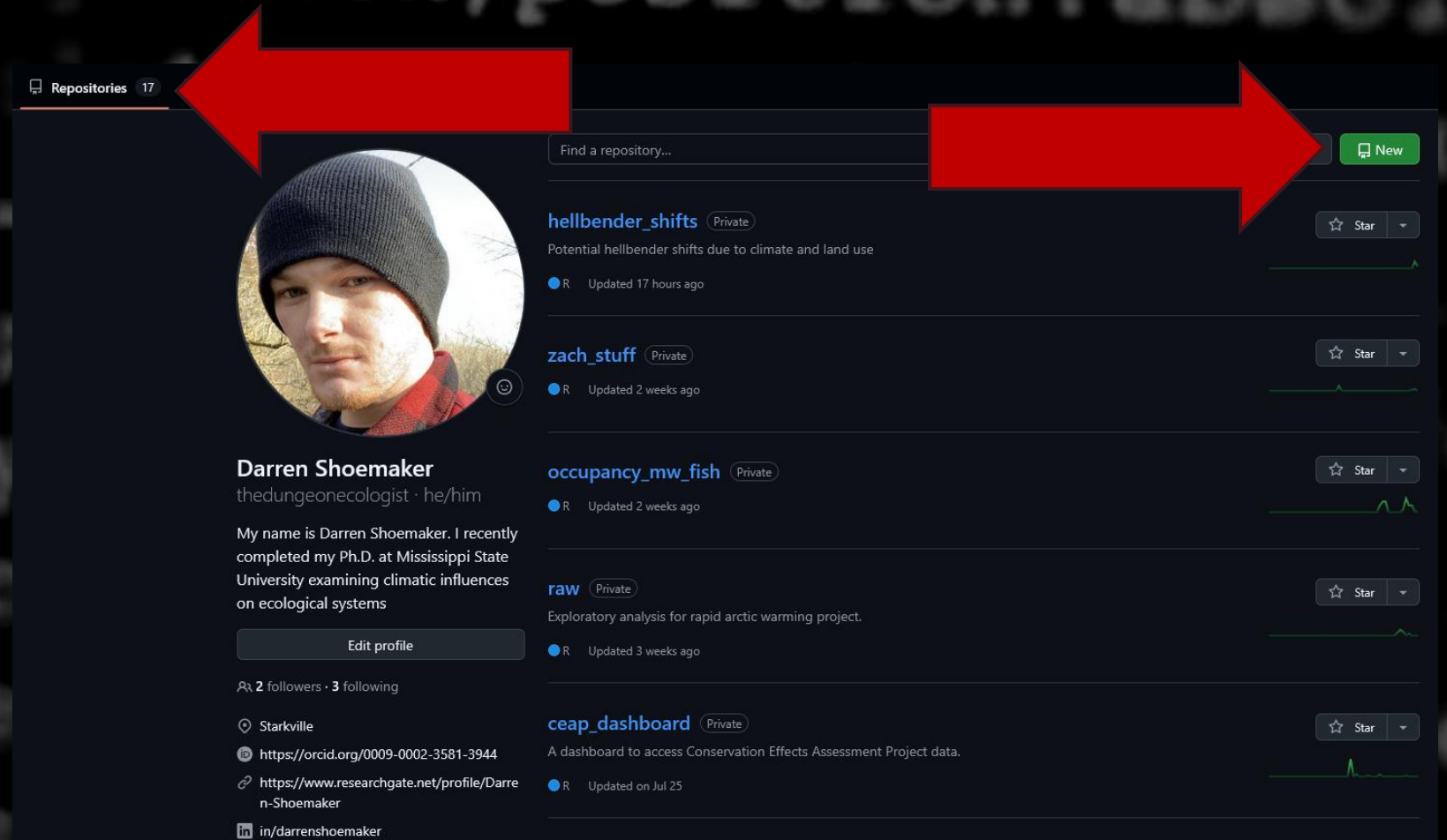
[Citing usethis](#)

DEVELOPERS

[Hadley Wickham](#)
Author 

CONNECT GIT TO GITHUB

- Make sure your computer can connect to GitHub
- This is a simple check from the Terminal
- Usually we will do this with **usethis**



CONNECT GIT TO GITHUB, CONT.

- Repos made from GitHub are named with a URL
- USERNAME/REPO_NAME
- Turn on the README
- .gitignore is a good habit
- Sidenote: What happens to my URLs if I change my username?

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * / Repository name *
thedungeonecologist / workshop
✓ workshop is available.

Great repository names are short and memorable. How about [automatic-computing-machine?](#)

Description
test for workshop
17 / 350 characters

2 Configuration

Choose visibility *
Choose who can see and commit to this repository
Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#)
On

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
R

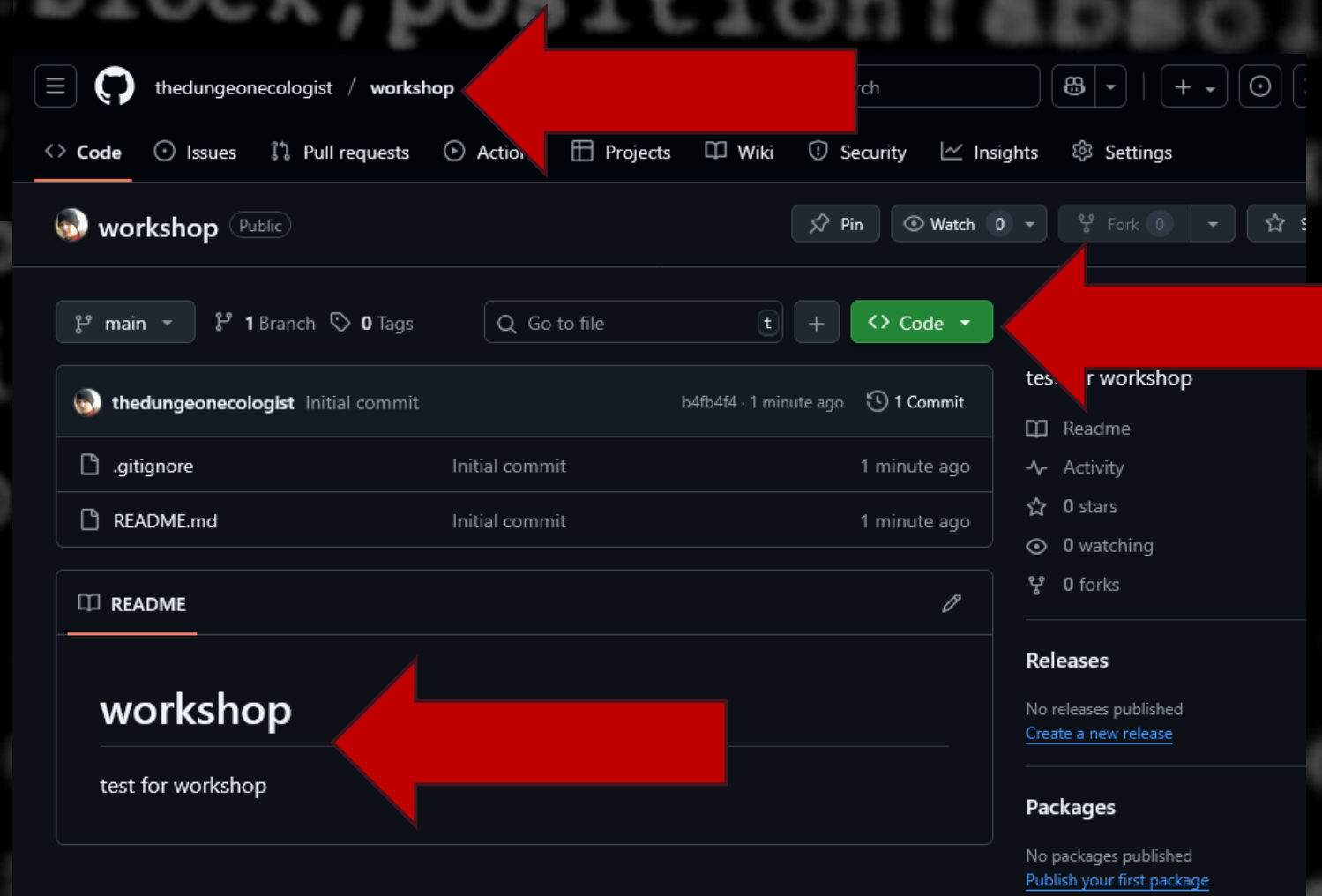
Add license
Licenses explain how others can use your code. [About licenses](#)
No license

Create repository

YOUR FIRST REPO

DO NOT BE INTIMIDATED!

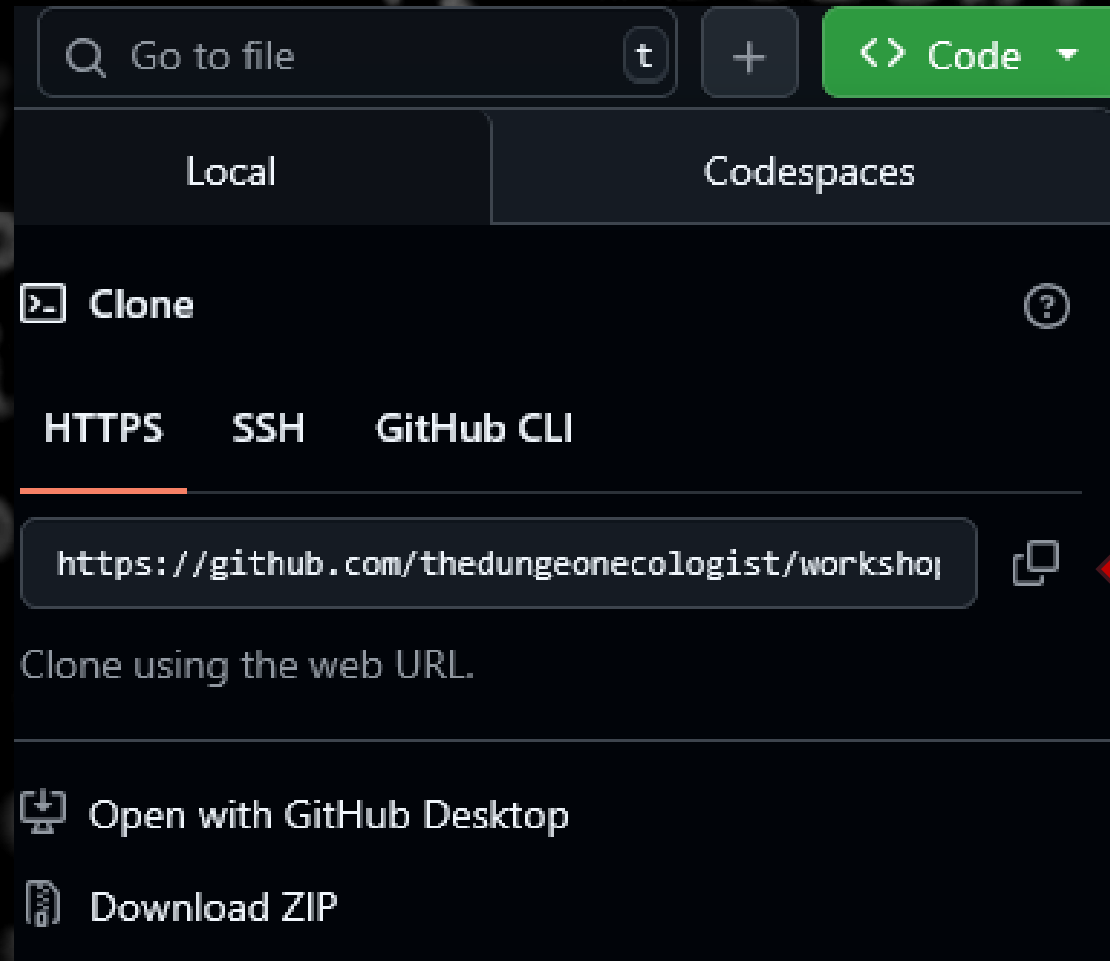
- We only care about a couple things here
- This is the URL for your repo, so you can get a sense of organization
- Here is how to access the files
- Here is the README you added



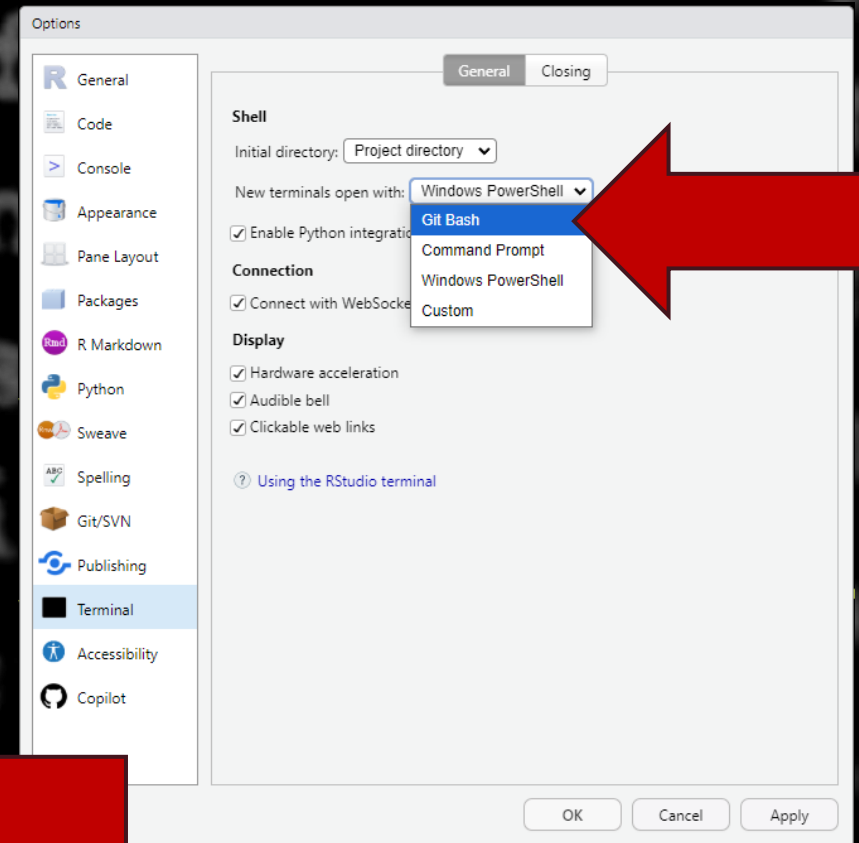
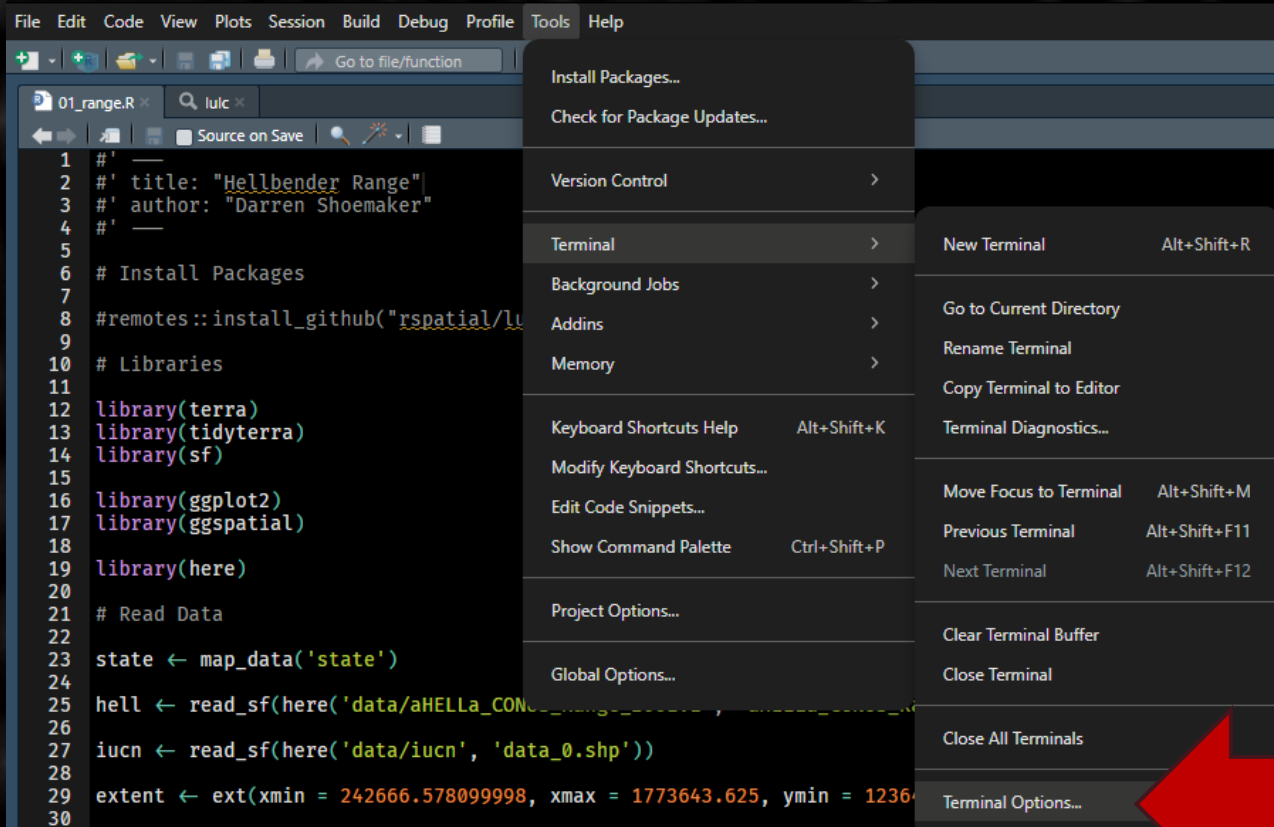
CAN YOU HEAR ME?

DO NOT BE INTIMIDATED!

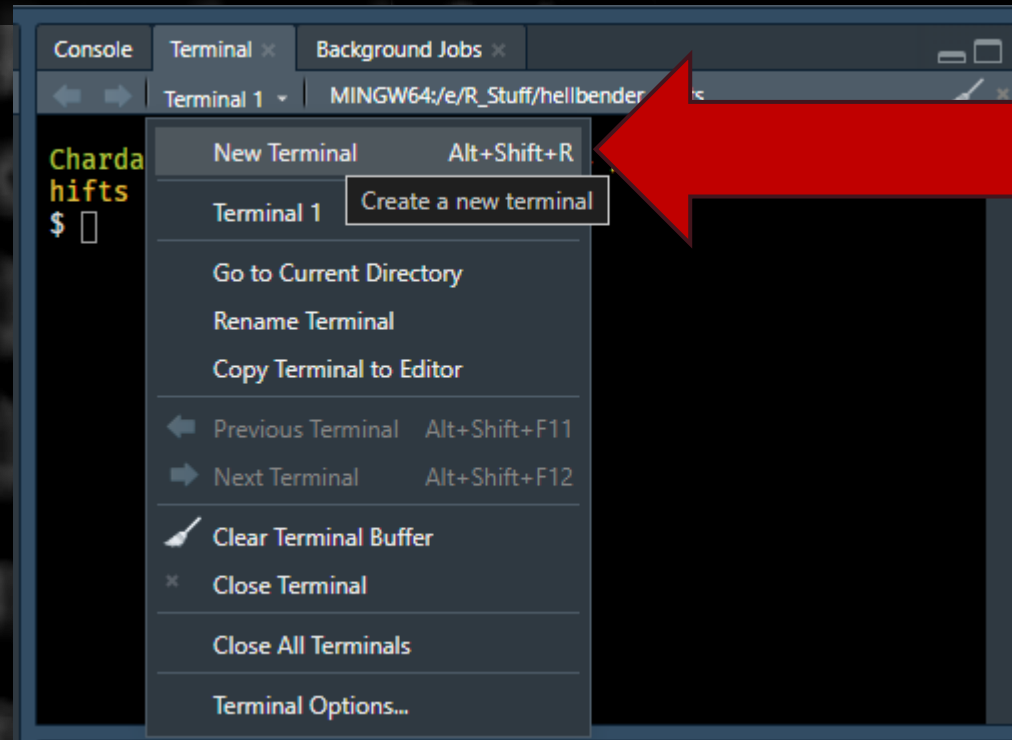
- Copy the URL
- Remember we used HTTPS
- That ZIP option at the bottom might be helpful if my friend doesn't use GitHub...



WHILE WE'RE HERE...

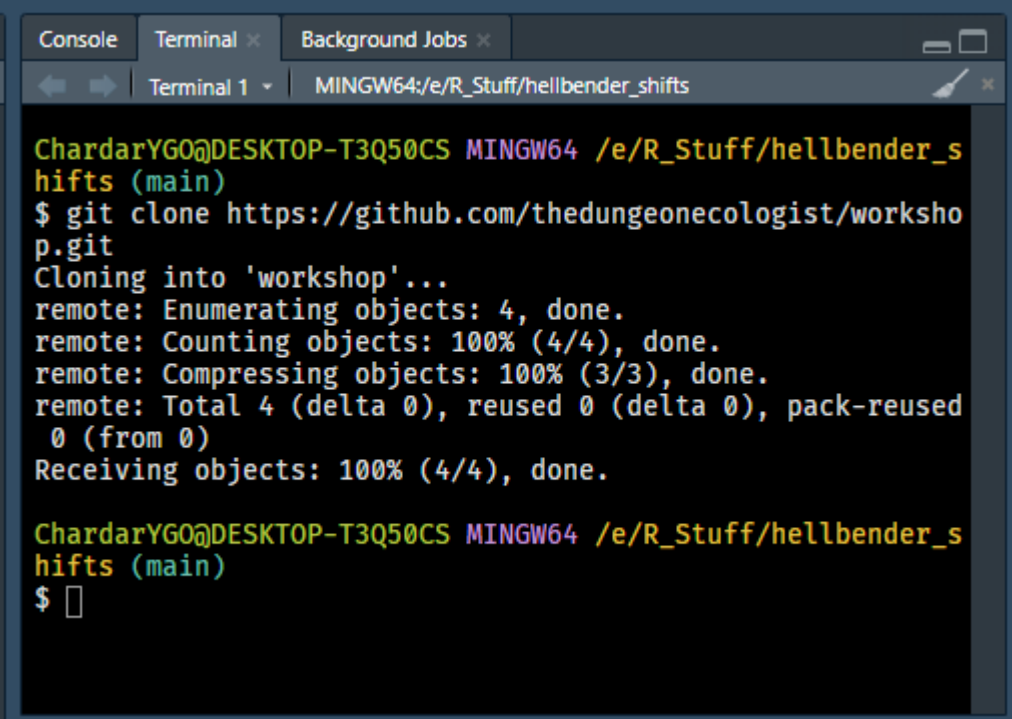


BACK TO THE TERMINAL



BACK TO RSTUDIO

- In the R console, note your working directory `getwd()`
- Your console should look a little different now because we are working in bash instead of shell
- You don't need to know what this means
- My takeaway from Stackexchange is no one knows what this means
- In the Terminal, type `git clone` and paste the URL from your repo

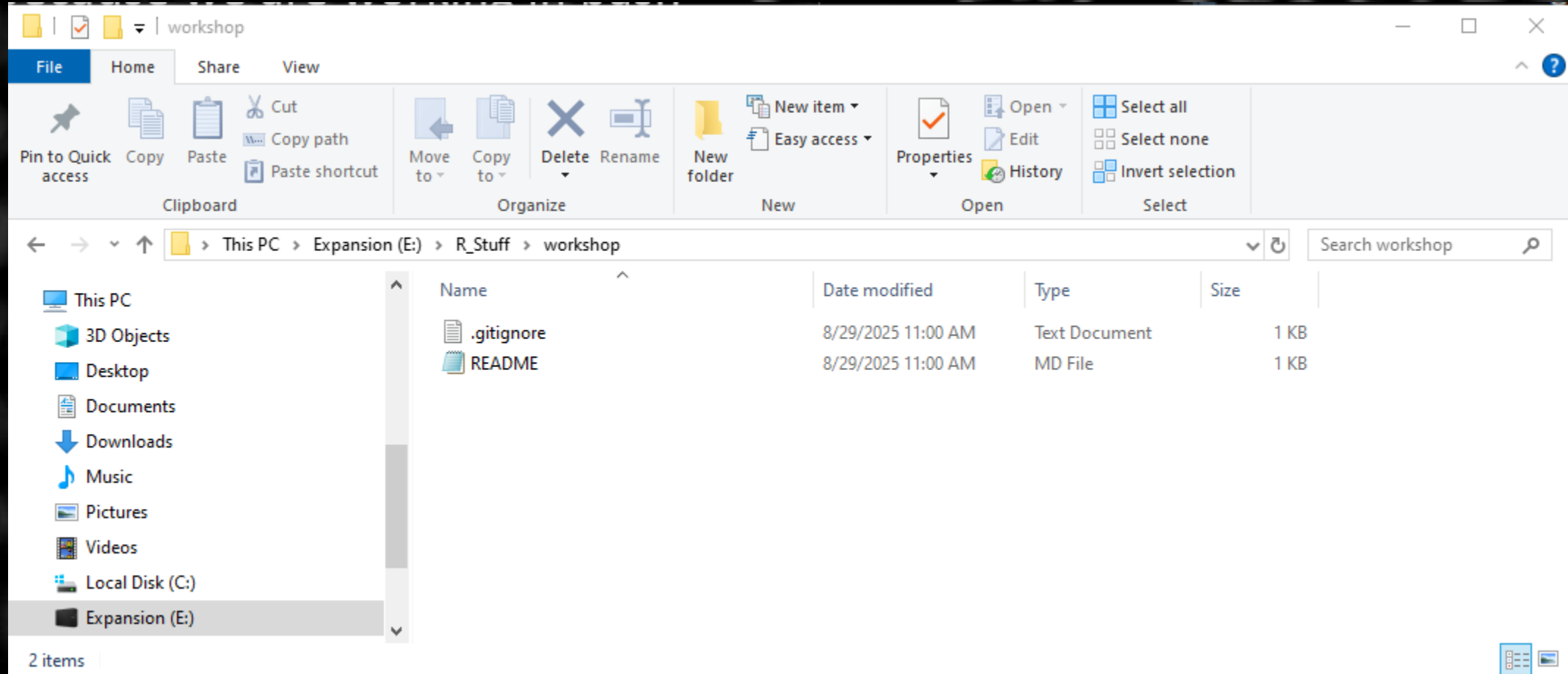


```
Console Terminal × Background Jobs ×
Terminal 1 MINGW64:/e/R_Stuff/hellbender_shifts

ChardarYG0@DESKTOP-T3Q50CS MINGW64 /e/R_Stuff/hellbender_s
hifts (main)
$ git clone https://github.com/thedungeonecologist/worksho
p.git
Cloning into 'workshop'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused
0 (from 0)
Receiving objects: 100% (4/4), done.

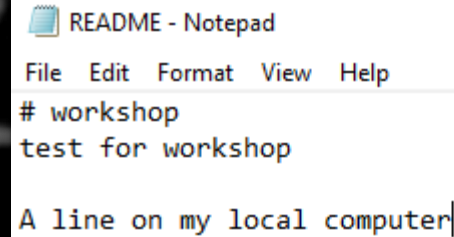
ChardarYG0@DESKTOP-T3Q50CS MINGW64 /e/R_Stuff/hellbender_s
hifts (main)
$
```

OH LOOK IT'S MY REPO!



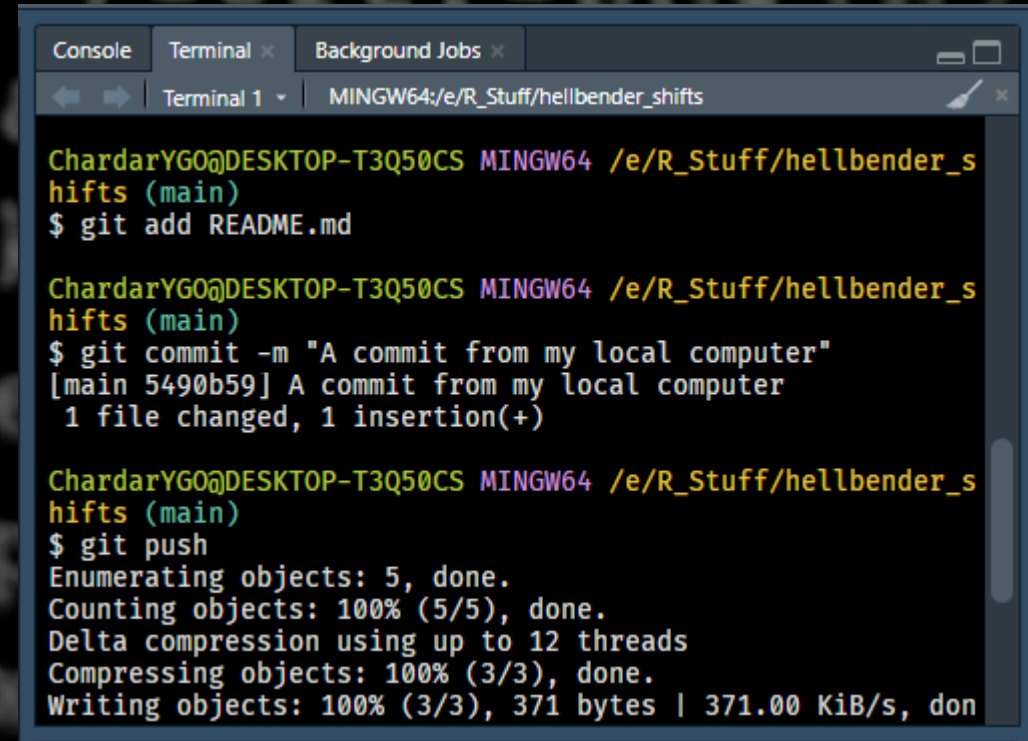
MAKE A CHANGE

- Open the README file in Notepad, add something, and save the file
- Go back to the Terminal
- Type `git add README.md`
- Type `git commit -m "A commit"`
- Type `git push`
- If asked for a password, use your PAT
- If asked for a username or email, use the same as your GitHub account



```
README - Notepad
File Edit Format View Help
# workshop
test for workshop

A line on my local computer|
```



```
Terminal 1 MINGW64:/e/R_Stuff/hellbender_shifts

ChardarYG0@DESKTOP-T3Q50CS MINGW64 /e/R_Stuff/hellbender_s
hifts (main)
$ git add README.md

ChardarYG0@DESKTOP-T3Q50CS MINGW64 /e/R_Stuff/hellbender_s
hifts (main)
$ git commit -m "A commit from my local computer"
[main 5490b59] A commit from my local computer
1 file changed, 1 insertion(+)

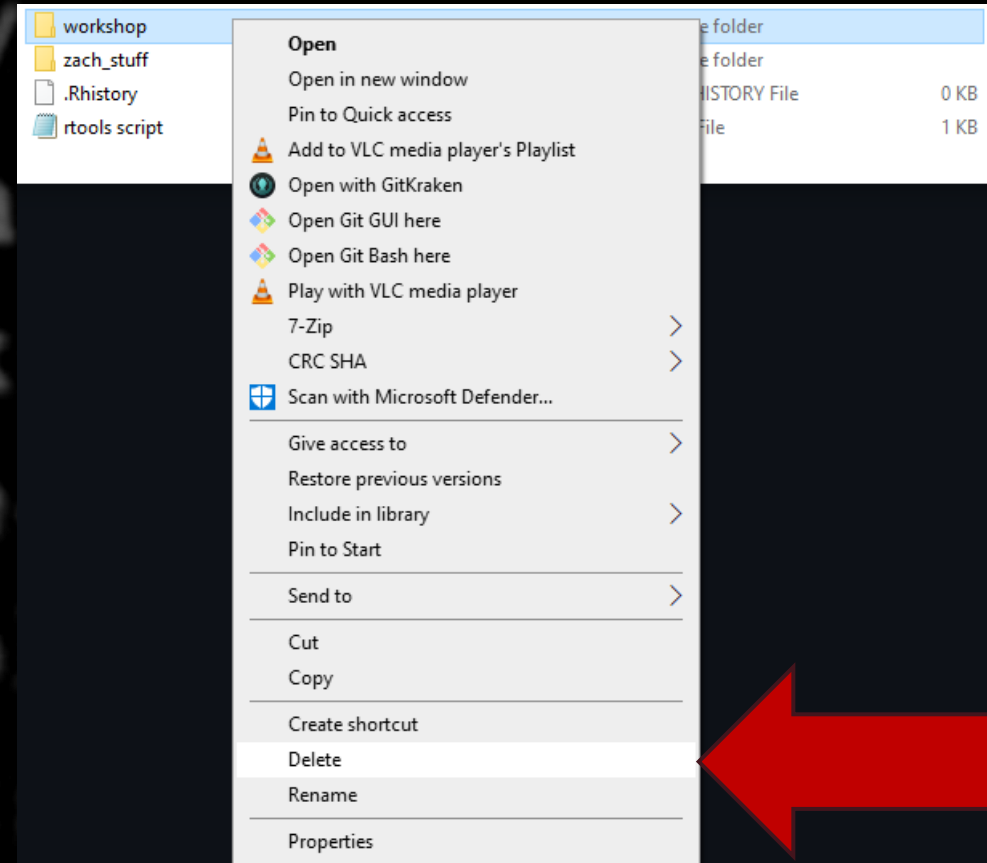
ChardarYG0@DESKTOP-T3Q50CS MINGW64 /e/R_Stuff/hellbender_s
hifts (main)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 371 bytes | 371.00 KiB/s, don
```

DID IT WORK?

- Got back to your repo on GitHub
- The change you made to the README file should appear online
- Hopefully that all worked
- If not, troubleshooting time!
- If that worked, delete the local folder (yes really)

README

A line I wrote on my local computer

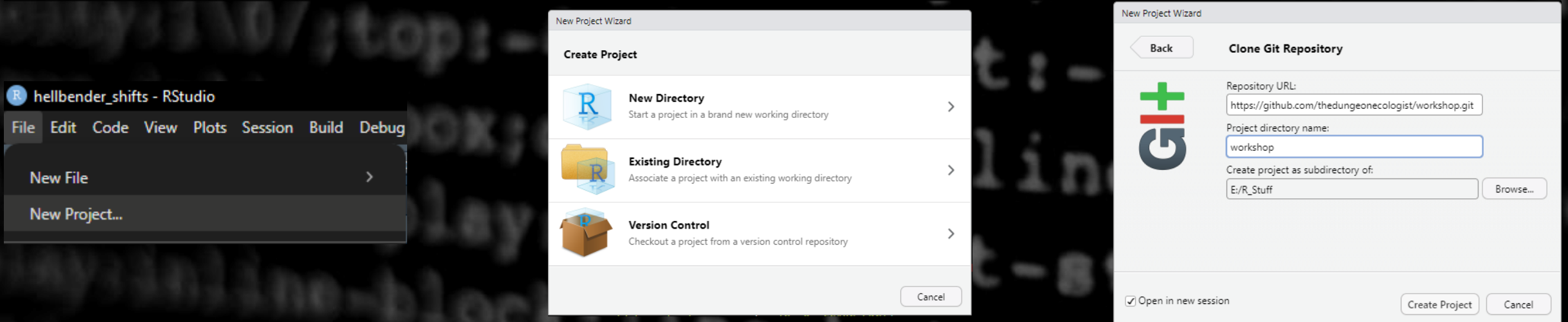


THE PART YOU ACTUALLY CAME FOR

Part 1: New R Project from GitHub

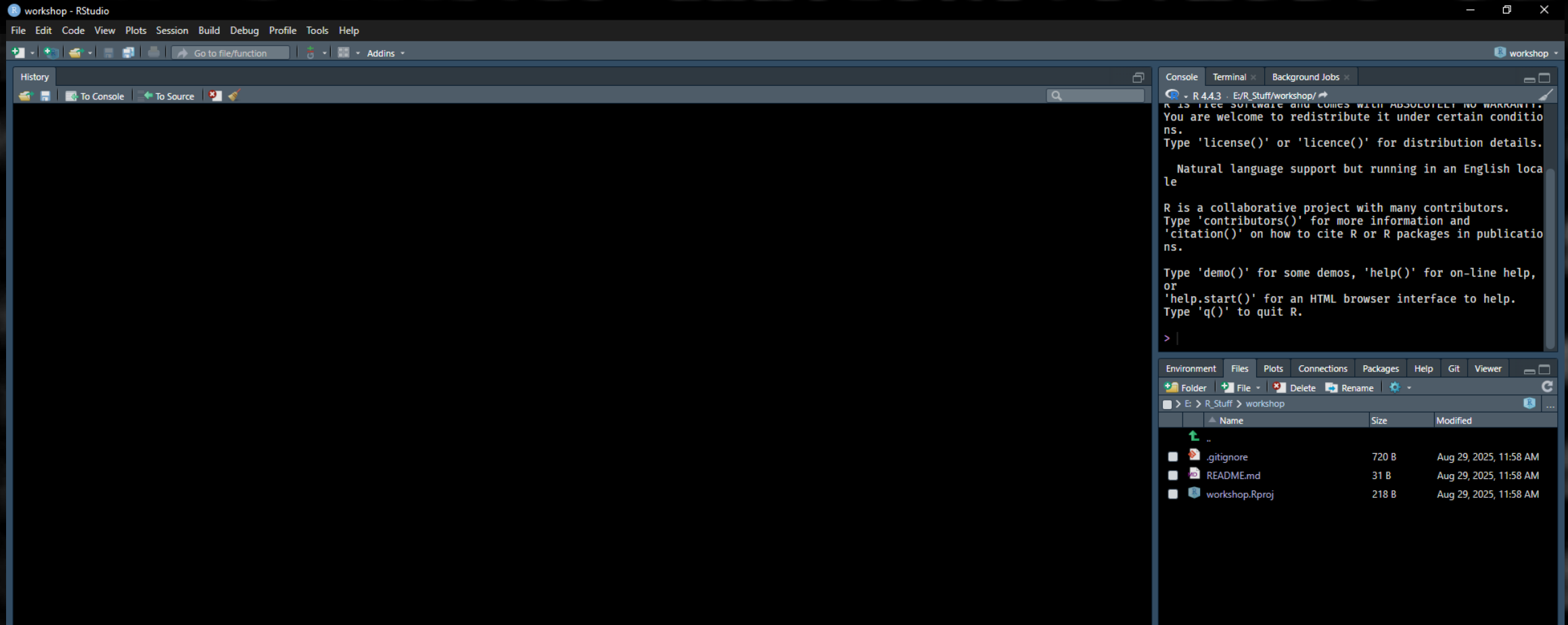
This is the recommended process

NEW R PROJECT FROM GITHUB

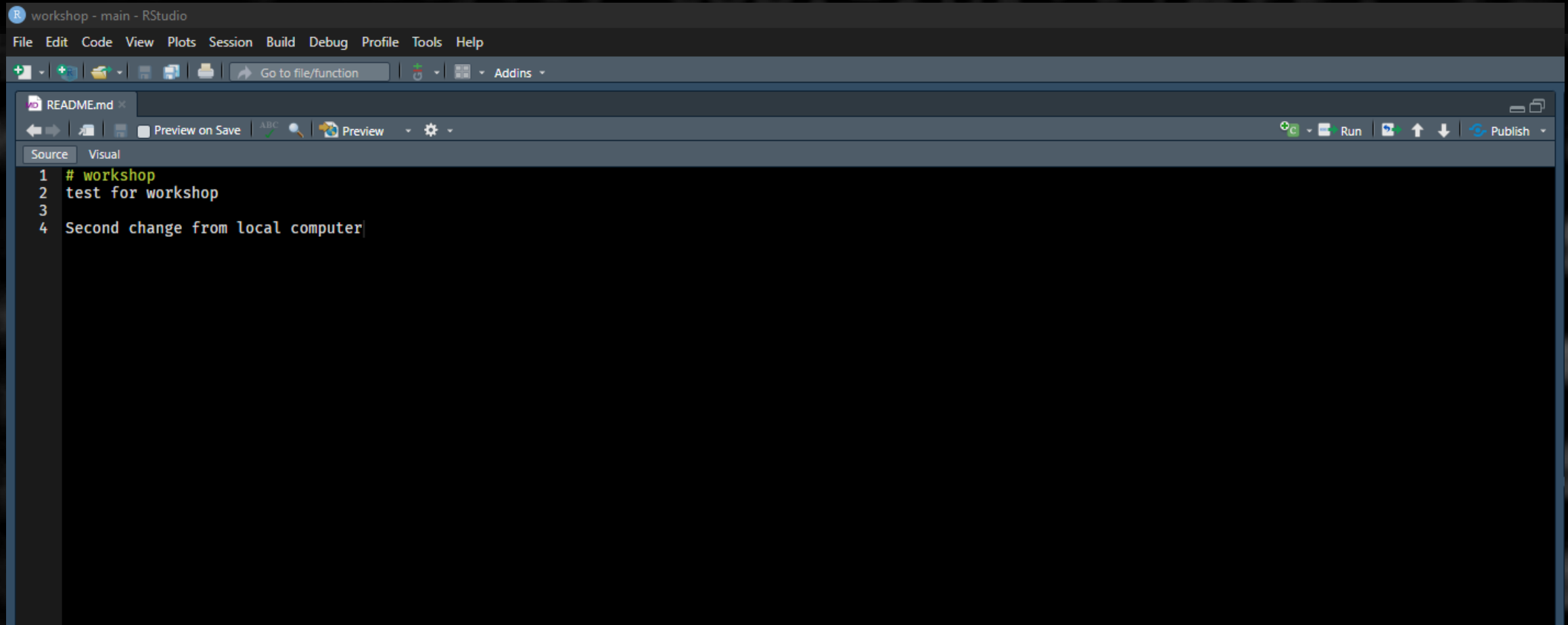


- Click version control -> click Git -> paste the URL from your repo
- Name the project directory (I recommend making this the same as the repo) -> check Open in New Session
- Click Create Project

ISN'T THIS WHAT I JUST DELETED?



MAKE AND SAVE ANOTHER CHANGE IN THE README



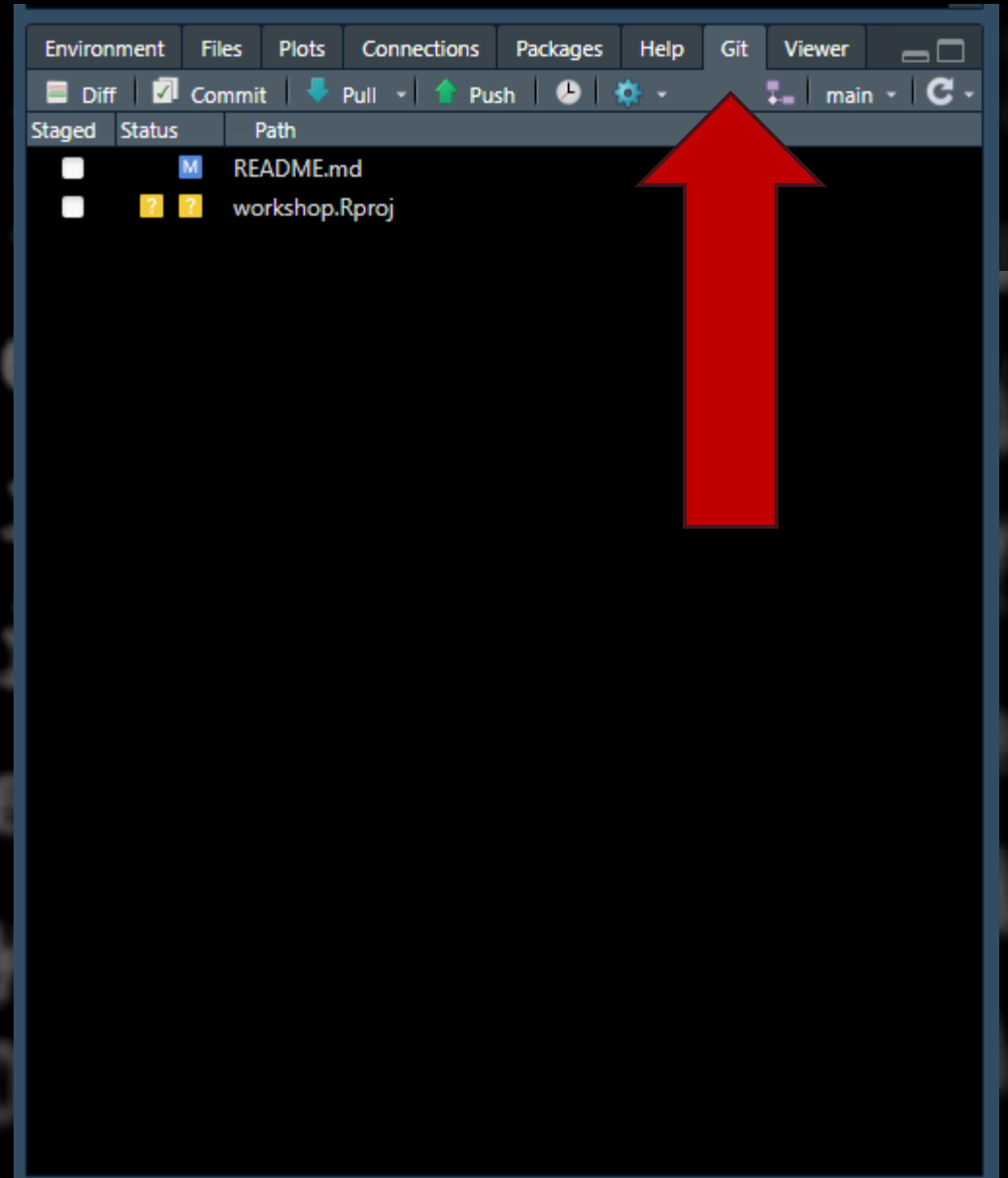
The screenshot shows the RStudio interface with a file named 'README.md' open. The editor has a 'Source' tab selected, showing the following content:

```
1 # workshop
2 test for workshop
3
4 Second change from local computer
```

The interface includes a menu bar (File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help), a toolbar with icons for file operations and a 'Go to file/function' search bar, and a status bar at the bottom with 'Preview on Save' and 'Preview' options. The right side of the window shows a 'Run' button and a 'Publish' button.

NAVIGATE TO THE GIT PANE

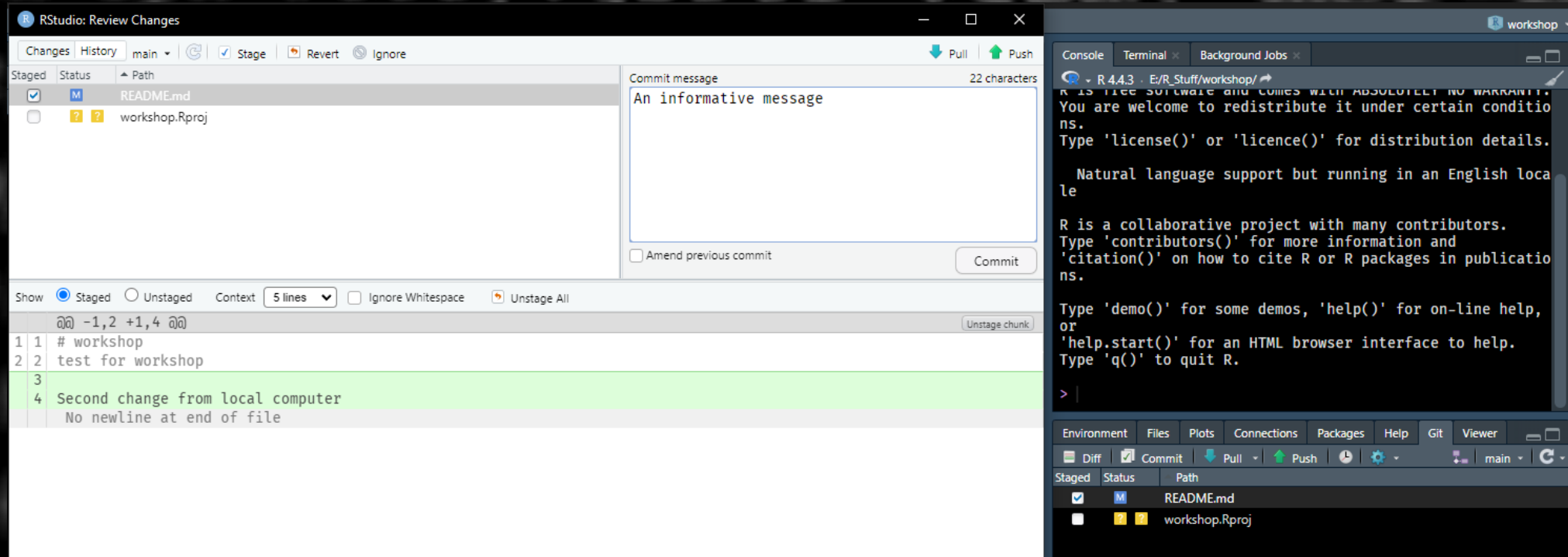
- Any time you save a change to a file in the R Project, it will appear here
- Small vocabulary detour ahead



BASIC GIT TERMS

- Diff – a record of changes made to a file
- Commit – a modification to a file you want Git to track
- Staged – a local commit that isn't on Github yet
- Push – send your commits to GitHub (main)
- Pull – get changes from GitHub (main) on your local machine

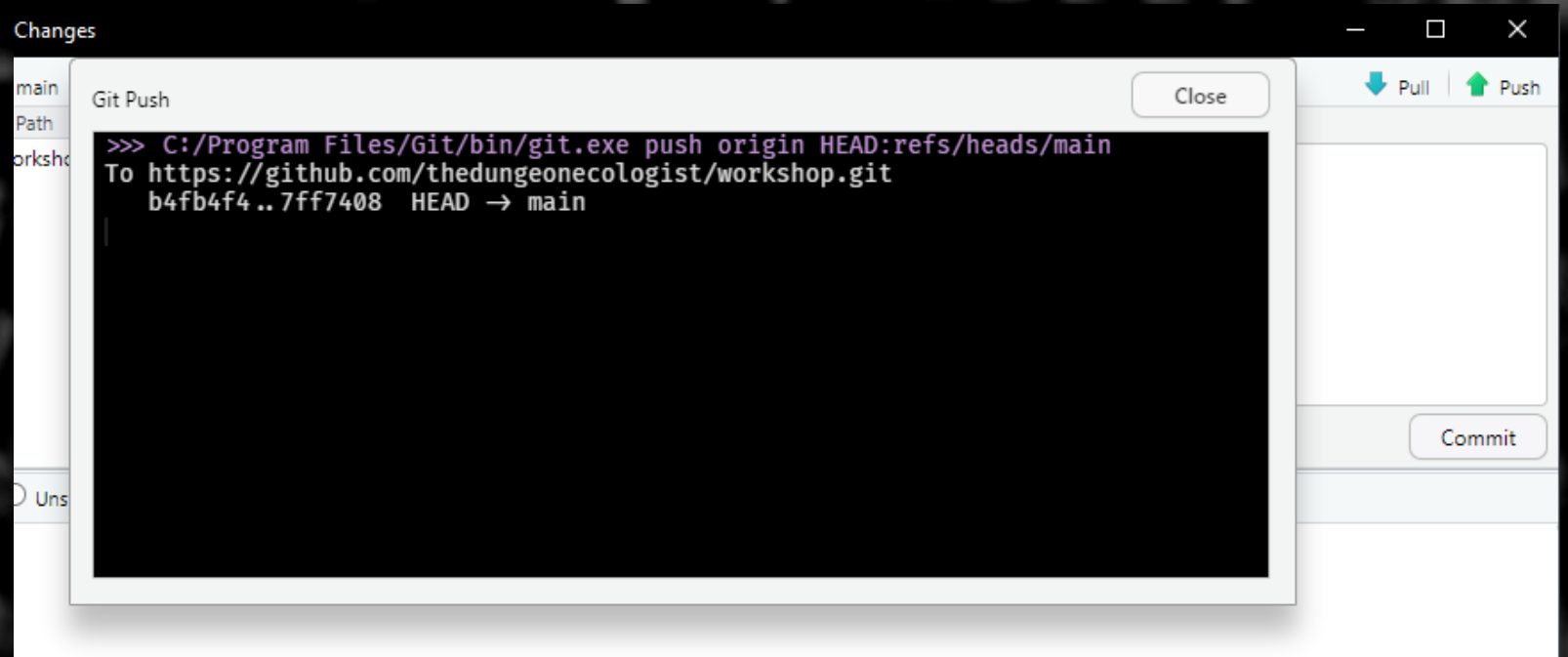
MAKE A COMMIT FROM RSTUDIO



- Check all the files you want stage
- Click commit in the Git pane to see the diffs
- Add an informative message (remember these are public)
- Click commit under the message
- Click push to send the changes to GitHub

MAKE A COMMIT FROM RSTUDIO

- If the push is successful, you should see something like this



MAKE A COMMIT FROM RSTUDIO

- Your changes and commit message should appear on the GitHub website

The screenshot displays a GitHub repository interface. At the top, the repository name 'thedungeonecologist' is shown with a profile picture, followed by the commit message 'An informative message', the commit hash '7ff7408', the time '6 minutes ago', and '2 Commits'. Below this, a table lists the files committed: '.gitignore' with the message 'Initial commit' and '1 hour ago', and 'README.md' with the message 'An informative message' and '6 minutes ago'. The 'README' file is selected, showing its content: 'workshop' as a section header, followed by 'test for workshop' and 'Second change from local computer'.

File	Commit Message	Time
.gitignore	Initial commit	1 hour ago
README.md	An informative message	6 minutes ago

README

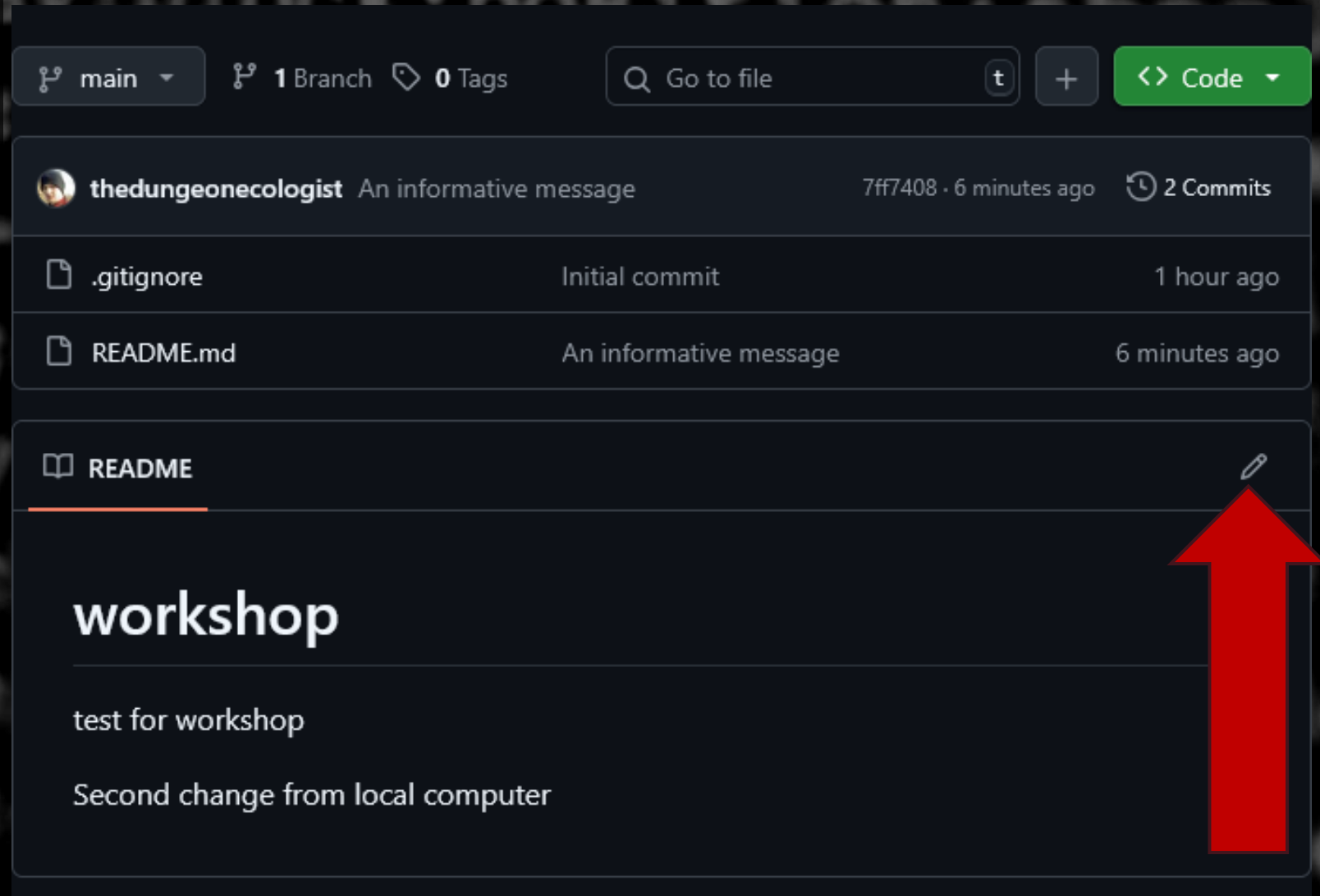
workshop

test for workshop

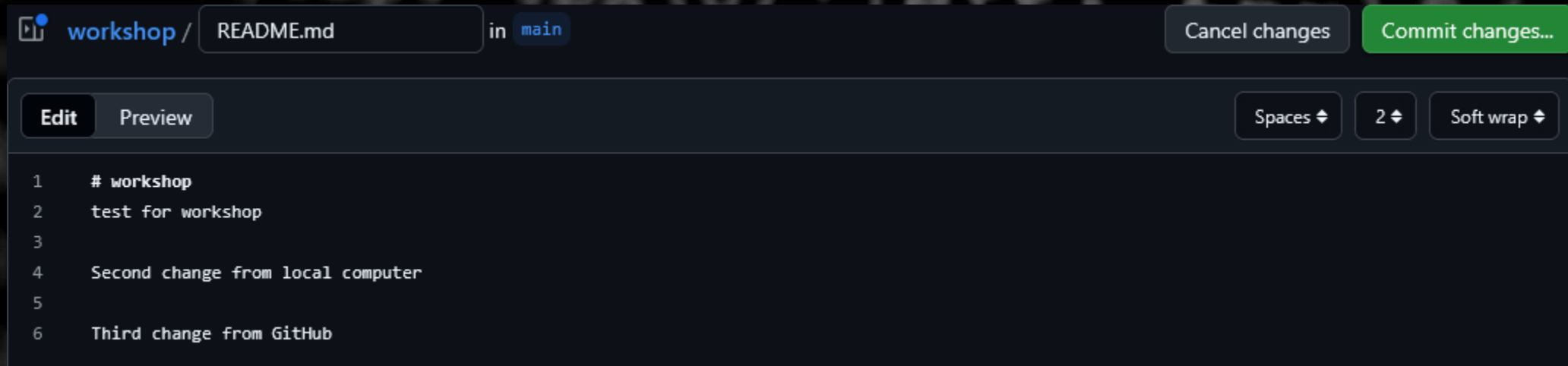
Second change from local computer

MAKE A COMMIT FROM GITHUB

- Edit the README from GitHub



MAKE A COMMIT FROM GITHUB



MAKE A COMMIT FROM GITHUB

- The default commit message should be fine
- Commit to the main branch for now

Commit changes

Commit message

Update README.md

Extended description

Add an optional extended description...

Commit Email

shoemaker.darrenj@gmail.com

☒ Commit directly to the main branch

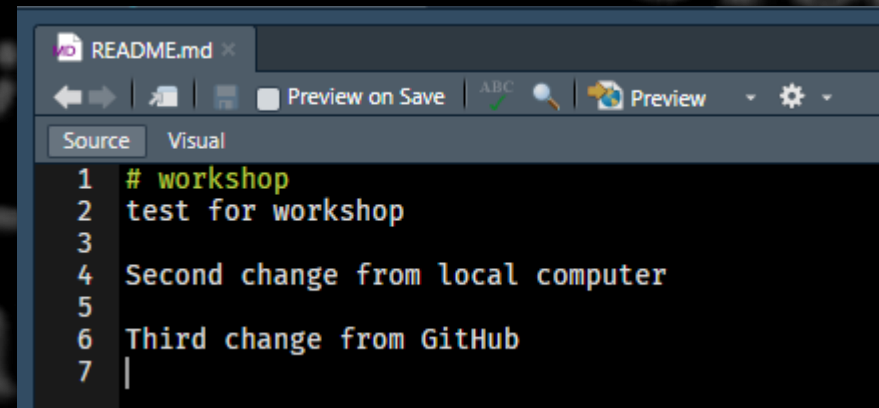
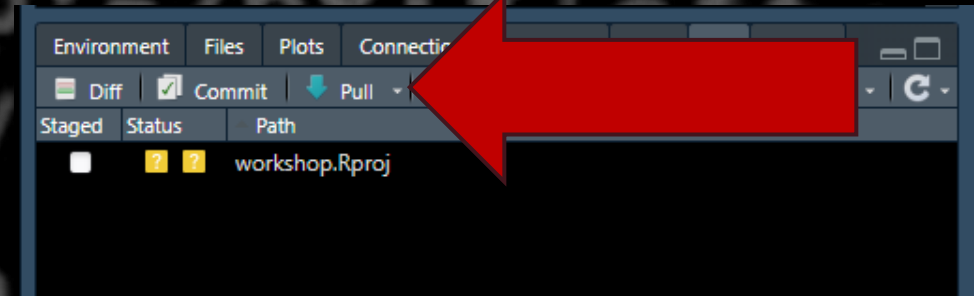
☐ Create a **new branch** for this commit and start a pull request [Learn more about pull requests](#)

Cancel

Commit changes

MAKE A COMMIT FROM GITHUB

- Back in Rstudio, click Pull
- Open the README to see your edit
- This process is the same no matter where the file was edited!
- This is how you can keep projects structured across devices and collaborators
- Make a habit of pulling at the start of your session to avoid merge conflicts



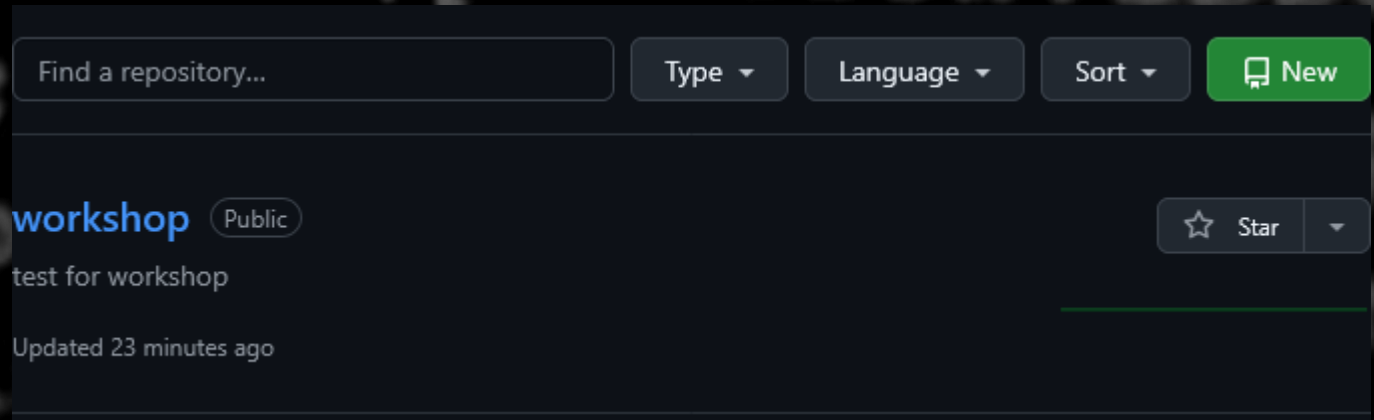
THE PART YOU ACTUALLY CAME FOR

Part 2: Moving existing work to
GitHub – GitHub first

You'll probably need this for a bit

NEW REPO FROM R PROJECT – GITHUB FIRST

- If your existing R work is not in a project, get it into one
- Go back to GitHub and make a new repo
- I recommend naming this the same as your project directory
- Copy the URL



Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk ().*

1 **General**

Owner * thedungeonecologist / Repository name *

Great repository names are short and memorable. How about `potential-succotash`?

Description

0 / 350 characters

2 **Configuration**

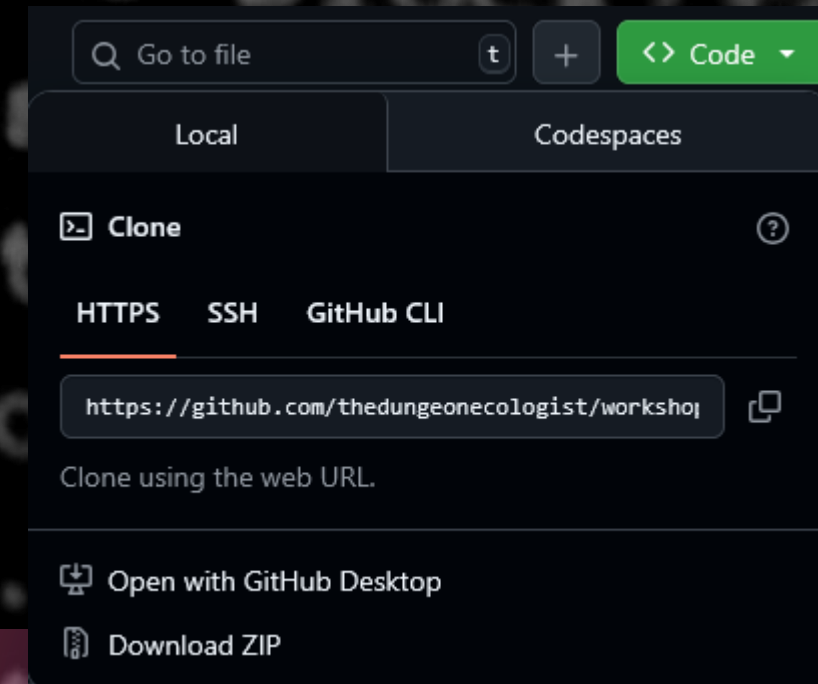
Choose visibility *
Choose who can see and commit to this repository Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

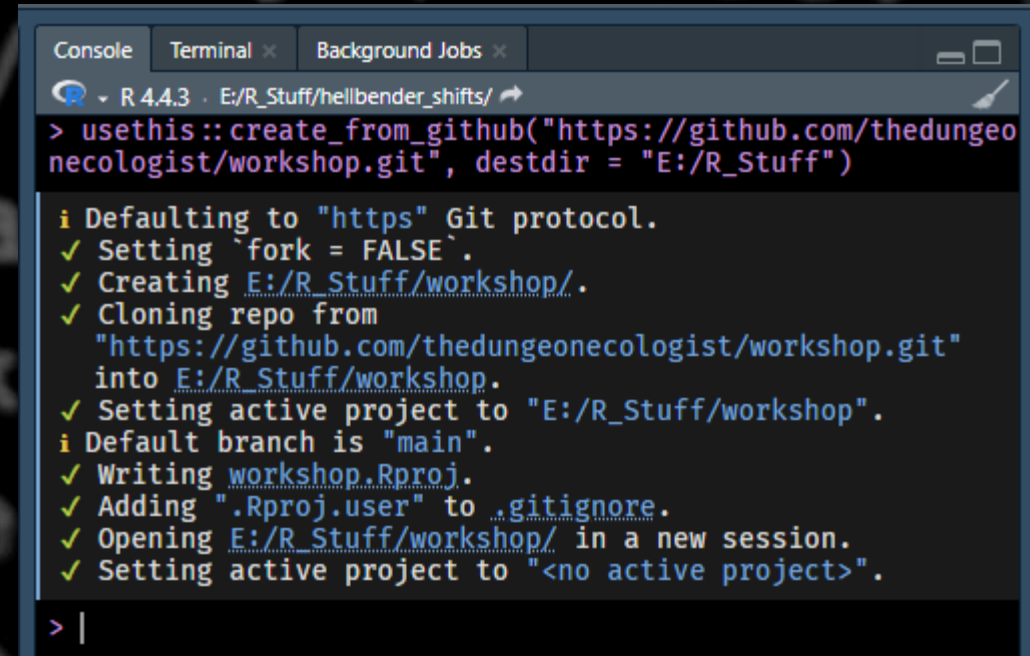
Add license
Licenses explain how others can use your code. [About licenses](#) No license

[Create repository](#)



NEW REPO FROM R PROJECT – GITHUB FIRST

- This can also be done with the new project tab like before, but I think this is simpler
- In the R console, type
`usethis::create_from_github("URL", destdir = "PATH_TO_DIRECTORY")`
- This will create a new project directory using the information from the GitHub repository



```
Console Terminal Background Jobs
R 4.4.3 E:/R_Stuff/hellbender_shifts/
> usethis::create_from_github("https://github.com/thedungeonecologist/workshop.git", destdir = "E:/R_Stuff")

i Defaulting to "https" Git protocol.
✓ Setting `fork = FALSE`.
✓ Creating E:/R_Stuff/workshop/.
✓ Cloning repo from
  "https://github.com/thedungeonecologist/workshop.git"
  into E:/R_Stuff/workshop.
✓ Setting active project to "E:/R_Stuff/workshop".
i Default branch is "main".
✓ Writing workshop.Rproj.
✓ Adding ".Rproj.user" to .gitignore.
✓ Opening E:/R_Stuff/workshop/ in a new session.
✓ Setting active project to "<no active project>".

> |
```

NEW REPO FROM R PROJECT – GITHUB FIRST

- Copy-paste the files you want associated with the GitHub repo to the new folder
- The new files will appear in the Git Pane
- Commit and Push like before

Expansion (E:) > R_Stuff > workshop

Name	Date modified	Type
.gitignore	8/29/2025 12:55 PM	Text Document
README	8/29/2025 12:55 PM	MD File
workshop	8/29/2025 12:55 PM	RStudio Project File
01_data_visualization	10/30/2023 12:23 PM	R File
00_R_Basics	10/18/2023 12:24 PM	R File

Environment	Files	Plots	Connections	Packages	Help	Git	Viewer
Diff	Commit	Pull	Push			main	
Staged	Status	Path					
☐	M	.gitignore					
☐	? ?	00_R_Basics.R					
☐	? ?	01_data_visualization.R					
☐	? ?	workshop.Rproj					

THE PART YOU ACTUALLY CAME FOR

Part 3: Moving existing work to
GitHub – GitHub last

Similar to part 2

NEW REPO FROM R PROJECT – GITHUB LAST

- Find an R project not associated with Git
- In the R console, type `usethis::use_git()`, commit, and restart when prompted
- Restart RStudio when prompted to activate the Git Pane
- Type `usethis::use_github()`
- The new repo should open in your browser
- Defaults are usually safe but you can call `?usethis::use_git()` and `?usethis::use_github()` if you want to change settings
- From here, push and pull just like before

```
> usethis::use_git()
✓ Setting active project to
  "E:/R_Stuff/retired/jeff".
✓ Initialising Git repo.
✓ Adding ".Rproj.user", ".Rhistory", ".RData",
  ".httr-oauth", ".DS_Store", and ".quarto" to
  .gitignore.
i There are 4 uncommitted files:
• .gitignore
• data/
• jeff.Rproj
• scripts/

! Is it ok to commit them?

1: Definitely
2: No
3: Nope

Selection: 1|
```

WHAT NOW?
WHY DID I DO ALL THIS?
PLEASE STOP

VERSION CONTROL

- Every file tracked on Git includes a History
- This shows the diff and commit messages for each date
- Collaborators can see when changes were made
- “Collaborators” includes you in a month
- You can go back to an older version if you break something

Commits

History for `res_clim` / `app.R` on `main`

All users

All time

Commits on Jul 9, 2025

Added headers to dynamic cards in UI. This functions similarly to the previous `uiOutput` but reduces the number of output objects required.

cac8fc9

thedungeonecologist committed on Jul 9

Updated additional reservoirs reset button to function like the study reservoirs one. Minor text changes to the study reservoirs table to make navigation more intuitive.

ab5954c

thedungeonecologist committed on Jul 9

Updated the study reservoirs tab reset button to work with `shinyjs::show/hide` functions instead of setting values to `NULL`. This performs as expected, clearing all outputs, resetting the values in t...

66201ed

thedungeonecologist committed on Jul 9

Updated `res_clim` to `ResClim` in frontend text per Corey Dunn's revisions. Also updated version number in YAML title.

1ce7906

thedungeonecologist committed on Jul 9

Commits on May 27, 2025

Removed ecoregion shapefiles which are now included in `res_long` database and associated `extract(ecoregion)` function in app. This should speed up the additional reservoirs tab.

ce755ab

thedungeonecologist committed on May 27

Commits on May 16, 2025

Integrated time series figures across years into downloadable report. Other minor text changes.

7daa924

thedungeonecologist committed on May 16

Commits on Apr 29, 2025

Added PRISM to references.

fae3b99

thedungeonecologist committed on Apr 29

Reformatted # comments to make code easier to navigate. Added PRISM to references.

0969fc6

thedungeonecologist committed on Apr 29

COLLABORATION

- You can invite collaborators to work on your repository
- They can pull your work and push their own modifications
- You have control over the main branch and can decide to approve their changes
- Federal agencies require* sharing code through Git for Fundamental Science Practices (FSP) review
- *Unless you work on vaccines, climate, gender...

Collaborators and teams



Private repository

Only those with access to this repository can view it

Manage visibility

Direct access



5 entities have access to this repository. [4 collaborators](#). [1 invitation](#).

Manage access

Add people

☐ Select all

Type ▾

Q Find a collaborator...



Caleb Aldridge

AldridgeCaleb • Collaborator



corey_dunn@ncsu.edu

Awaiting response

Pending Invite

Remove



fumosoftware

fumosoftware • Collaborator



kruppert

Collaborator



smiranda2

Collaborator



SHARING YOUR WORK

- GitHub is a sharable archive of your work
- You can add R scripts, spreadsheets, HTML documents
- Your work is reproducible
- You now have a portfolio that demonstrates your skills

The screenshot shows a GitHub repository named 'res_pollution' by user 'thedungeonecologist'. The repository is public and has 1 branch and 0 tags. The file list includes 'data', 'scripts', 'temp', '.gitignore', 'README.md', and 'res_pollution.Rproj'. The README file is selected, showing the title 'Assessment of pollution and other impairments of US reservoirs based on expert opinion'. The README content describes the repository's purpose, the analyses performed, and provides citation information for a paper by Miranda LE, Shoemaker DJ, and Krogman RM (2024).

Repository Details:

- Repository: `res_pollution` (Public)
- Branches: 1 Branch
- Tags: 0 Tags
- Commits: 4
- Stars: 0
- Forks: 0

File List:

File	Commit	Time
data	Initial commit	9 months ago
scripts	Initial commit	9 months ago
temp	Initial commit	9 months ago
.gitignore	Update gitignore	9 months ago
README.md	Create README.md	9 months ago
res_pollution.Rproj	Initial commit	9 months ago

README Content:

Assessment of pollution and other impairments of US reservoirs based on expert opinion

This repo contains preliminary analyses associated with the paper Assessment of pollution and other impairments of US reservoirs based on expert opinion.

Analyses performed: pairwise permanova, canonical analysis of principal coordinates (CAP), cluster analysis by Gower dissimilarity, multinomial regression

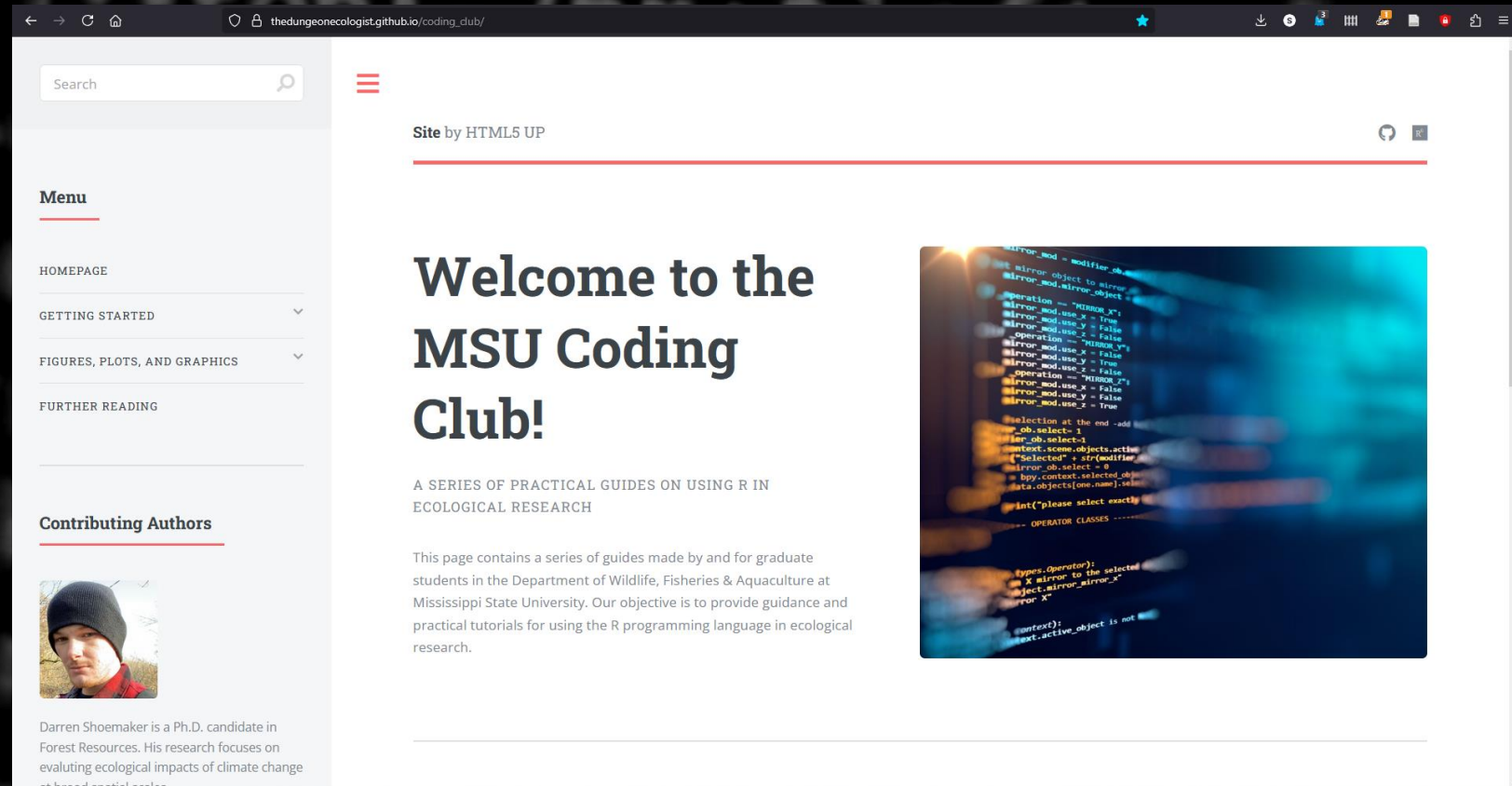
Please cite the manuscript as follows:

Miranda LE, Shoemaker DJ and Krogman RM (2024) Assessment of pollution and other impairments of US reservoirs based on expert opinion. Front. Environ. Sci. 12:1488955. doi: 10.3389/fenvs.2024.1488955

@article{miranda2024assessment, title={Assessment of pollution and other impairments of US reservoirs based on expert opinion}, author={Miranda, Leandro E and Shoemaker, DJ and Krogman, RM}, journal={Frontiers in Environmental Science}, volume={12}, pages={1488955}, year={2024}, publisher={Frontiers Media SA} }

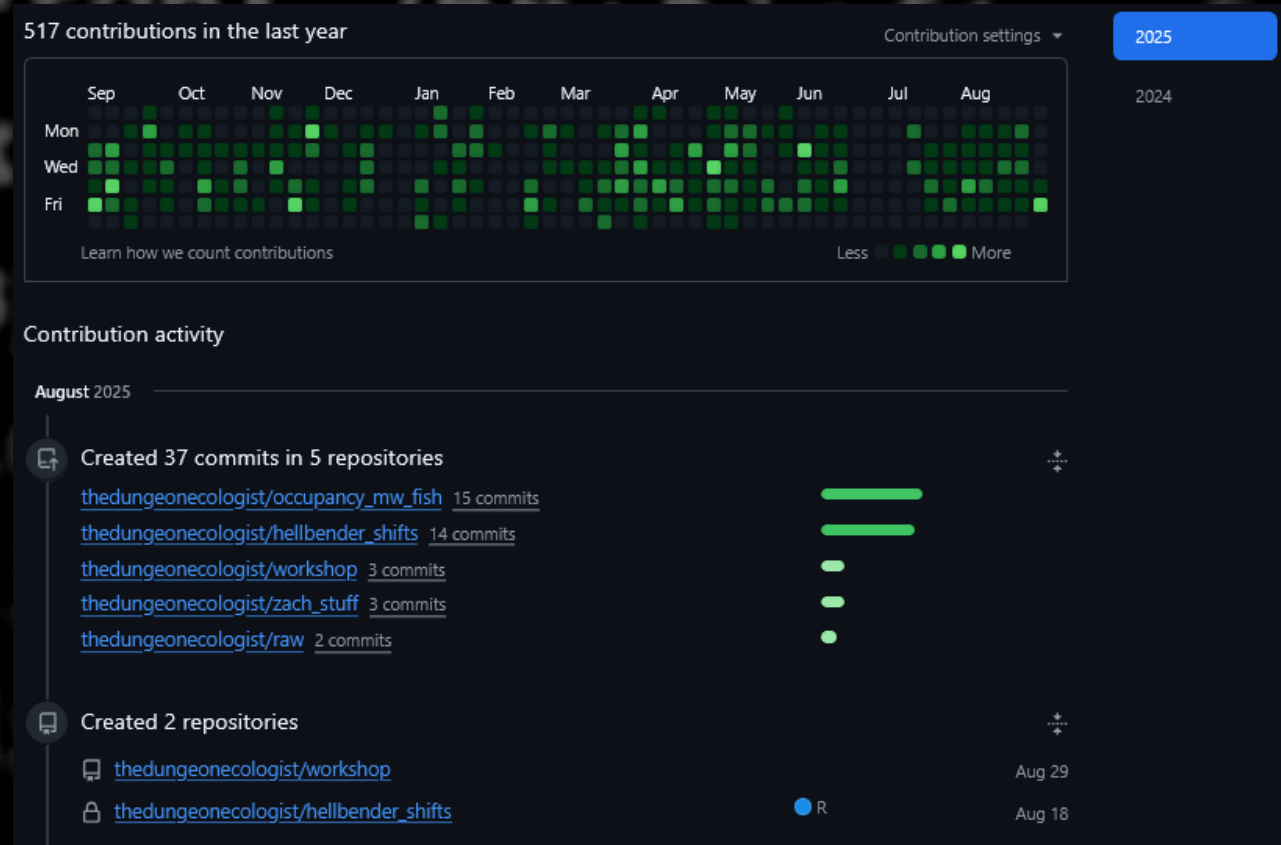
YOUR PERSONAL WEBSITE

- GitHub can host a website through HTML documents



TRACKING COMMITTS

- GitHub displays your commits on your profile
- Computer science recruiters go crazy for this
- This is dumb
- My boss is really patient



WRAPPING UP

DO NOT!

- Use spaces in any paths, folder, or file names
- Make a GitHub repo inside of another repo
- Try to push large files to GitHub
- Files should not be larger than 50mb and entire repos cannot be larger than 2gb
- Push sensitive or proprietary data to public repos
- Try to do everything – no one person will ever need the entire range of Git functions
- Give up



Questions?

djshoema@buffalo.edu

- This was an extremely basic overview of Git and GitHub with RStudio
- GitHub has a lot of functionality I've never tried
- If you are interested in testing a collaborative feature, let me know
- Feel free to reach out with any questions!

ResearchGate



TROUBLESHOOTING

MAKE SURE RSTUDIO KNOWS WHERE GIT IS

- This usually “just works”
- If the Git Pane does not appear in RStudio, check a few things in Global Options
- Is Version Control enabled?
- Can RStudio find the Git executable? (this is the Git file we installed awhile ago)
- Make sure cmd/ does not appear in the path
- Make sure this is set to git.exe, not git-bash.exe or a similar file
- Restart RStudio

